

Linear Order in the Language Faculty

Richard S. Kayne
New York University

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Part I. Background.

1. Word-order restrictions.

That human languages differ in their word order is a well-known fact. A simple example is:

- (1) You know them.
- (2) Tu les connais. ('you them know')

In English, the object pronoun follows the verb. In the corresponding French example in (2), the object pronoun *les* precedes the verb. A very large number of such word order differences are readily found across the world's languages.

A question immediately arises. Are there restrictions on possible word orders, or is any imaginable word order possible? The answer to this question, with virtual certainty, is yes, there are significant restrictions. Coordination provides one example, as argued by Zwart (2009).¹ According to Zwart, if one looks cross-linguistically at NP-coordination counterparts of English *and*, and if one limits oneself to coordinations in which this counterpart of *and* appears only once, one finds that *and* and its counterparts invariably occur between the two conjuncts:

- (3) a. NP and NP
- b. *and NP NP
- c. *NP NP and

Though they might seem perfectly plausible word orders, neither that of (3b) nor that of (3c) is found.

A more intricate cross-linguistic word-order restriction was discovered by Greenberg (1966) in his study of DP-internal word order.² Greenberg found that if you look at the pair demonstrative (Dem) and noun (N), you find that both ordering possibilities (Dem N, as well as N Dem) are attested cross-linguistically. And the same holds for numeral (Num) and N, with Num N order attested in some languages, and N Num order in others. And similarly for the relative order of adjective (Adj) and noun, insofar as both Adj N and N Adj orders are attested.

Looking at the above three pairs separately gives the initially misleading impression that the language faculty is word-order-wise extremely flexible, with few word order restrictions. Yet Greenberg showed in his Universal 20 that, if instead of examining word order through pairs of categories, you examine combinations in which all four of these categories (N, Dem, Num, Adj) occur together, the picture changes considerably. An asymmetry suddenly appears, as follows. If N is final, then the order of the other three is fixed, and one has only 'Dem Num Adj N' order. Whereas if N is initial the order of the other three is not fixed.

2. Antisymmetry.

Further questions arise. Why should there be such word order restrictions? And why these particular ones?

Zwart draws from (3) the reasonable conclusion that the limitation in question must be reflecting the absence of any syntactic movement. In terms of Chomsky's (1970) X-bar theory, this seems to indicate that *and* is a head (H),³ with the two conjuncts then

¹For additional relevant discussion of coordination, cf. Kayne (1994, chap. 6) and Zhang (2009; 2023), as well as references cited in those works.

²Whether a phrase such as *the book* is a NP, with the noun *book* as its head, or a DP, with the determiner *the* as its head, is an important question that will be touched on later in section 24 on definite articles, but is not central here.

³Note in particular Oda's (2017, sect. 5) taking Japanese &P to be head-initial.

The exact implications of Kayne (2016) for the general conclusion about *and* and its counterparts need to be determined, relevant to which is the fact that French allows, in addition to:

- i) ou Marie ou Jean ('or M or J')

the use of the more complex *ou bien*:

- ii) ou bien Marie ou bien Jean ('or well M or well J')

being specifier (S) and complement (C) of *and*. In which case (3a) is displaying either S-H-C order or C-H-S order. More carefully put, since H and C form a constituent to the exclusion of S, (3a) must correspond to one or another of the following:

(4) S [H C]

(5) [C H] S

There is empirical evidence that (4) is correct and that (5) is not correct. Consider first the following binding asymmetry involving the bound reading of *his*:

(6) ?Every little boy and his mother came to the party.

(7) *His mother and every little boy came to the party.

which suggests that the first conjunct c-commands the second, but not vice versa. This supports (4).

Second, we can note the following:

(8) They went to the store and bought food.

(9) They bought food and went to the store.

The first of these has a very natural interpretation that is temporally asymmetric, with the *going to the store* leading to the *buying of food*. That same interpretation is absent from the second example (in a way that would be surprising if coordination were symmetric). As to why this asymmetry should hold, let me suggest a link to:

(10) They went to the store and then bought food.

(11) They bought food and then went to the store.

the idea being in part that (8) contains a silent counterpart THEN (I will use capitals to indicate silence) of the overt *then* seen in (10). This is shown for (8) in:

(12) they went to the store and THEN bought food

Furthermore, both *then* and THEN, when contained in one conjunct, can take the other conjunct (or a subpart of it) as antecedent, but only if that other conjunct c-commands the conjunct in which *then* or THEN is located, as in fact holds in (8)/(12).

The reason that (9) lacks the interpretation of (8)/(12) can now be understood as follows. If THEN appears within the second conjunct in (9), the interpretation must rest on THEN taking (part of) the first conjunct as antecedent, in which case (9) will have the interpretation of (11). For (9) to have the interpretation of (8)/(12), it would need to have THEN within the first conjunct taking the second conjunct (or part of it) as antecedent. But that is impossible, in a Condition C-like way,⁴ if the first conjunct asymmetrically c-commands the second conjunct, which is exactly what we had already concluded based on (6) vs. (7).

Put another way, the preceding facts about coordination are to be understood in terms of (4), repeated here:

(13) S [H C]

with the specifier S (the first conjunct) asymmetrically c-commanding the complement C (the second conjunct).

The next question is, why should (13) hold of coordination? The answer given in Kayne (1994) is straightforward. Syntax is antisymmetric. (13) holds of coordination because (13) holds of all phrases. For all phrases, the specifier S that asymmetrically

⁴On a possible reduction of Condition C of the Binding Theory to a ban on downward movement, see Kayne (2002).

c-commands the constituent [H C] invariably precedes [H C], in all languages, as a core property of the language faculty.

Put somewhat differently, antisymmetry as in Kayne (1994) has it that asymmetric c-command invariably maps to precedence, without exception. There is no optionality of the sort associated with the head parameter approach to word order that was common in the 1980s. In effect, then, we have:

(14) Word order is entirely determined by hierarchical structure.

Let us now recall briefly Greenberg's generalization concerning DP-internal word order touched on at the end of section 1: When N, Dem, Num and Adj cooccur in a single phrase, their relative order, if N is final, is 'Dem Num Adj N'. If N is initial, their relative order is less fixed.

As Cinque has shown in papers and in a monograph,⁵ this asymmetry between fixed prenominal order and less fixed postnominal order fits directly into the 1994 antisymmetry proposal of mine just mentioned, from the perspective of which such left-right (pre-N/post-N) asymmetries are in a general way expected. As Cinque (2009; 2023) further shows, such complex left-right (pre-X/post-X) asymmetries are in fact more widely found than just in the case corresponding to Greenberg's Universal 20. (And as in Greenberg's case, these asymmetries come to the fore only if one examines the relative order of larger sets, not just pairs, of elements.)

For 'Dem Num Adj N', Cinque's proposal is, more specifically, that the external merge order (base order, in earlier terms) must have N asymmetrically c-commanded by Adj, in turn asymmetrically c-commanded by Num, with Dem then being externally merged in the relatively highest position. If no movements intervene, this array of asymmetric c-command relations maps directly to the order 'Dem Num Adj N' (since asymmetric c-command always maps to precedence).⁶

3. Word order and morpheme order.

Kayne (1994, sect. 4.5) took antisymmetry to be relevant below the word level, too. This amounted to saying the following, in a way parallel to (14):

(15) Morpheme order is entirely determined by hierarchical structure.

If we take the term 'linear order' that appears in the title of this essay to cover both word order and morpheme order, then we can subsume (14) and (15) under:⁷

(16) Linear order is entirely determined by hierarchical structure.

That a general principle like (16) should cover both words and morphemes fits directly into any framework that incorporates the reduction of morphology to syntax, as proposed in its strongest form in Collins and Kayne (2023).⁸

We can note in passing that the term 'morpheme order' would seem to be quite straightforward, even if, as is well-known, it is not always easy to delineate morpheme boundaries. The familiar term 'word order' is, despite its familiarity, arguably on less

⁵For details, see Cinque (2005; 2020a; 2023).

⁶For Cinque, leftward DP-internal movements are available to a limited extent, but must always involve N; consequently any movement operation will leave N non-final.

⁷On whether this should be extended to phonology, see the Appendix.

⁸Cf. Chomsky (1957) on affix-hopping, Baker (1985; 1988), Halle and Marantz (1993), Julien (2002) and Bruening (2018). On prefixes vs. suffixes, cf. Kayne (2017a).

solid ground, to the extent that the syntactic status of the notion 'word' is itself less solid. I have in mind in part Myers's (1987) argument that the orthographic word in the Bantu language Shona does not reflect any syntactic constituent, but only a phonological one. Related to this is Koopman and Szabolcsi (2000), who (p.103) "do not assume that all the components of a word must occur within a single phrase" and who (p.223) "are not assuming that the material that counts as a word for the purposes of phonology is dominated by any particular kind of syntactic label", as well as Julien (2002), who argues extensively that (p.3) "the notion of 'word' itself cannot be defined in syntactic terms". We might also think of the relative clause analysis of derived nominals developed in Kayne (2008a),⁹ which has the property that, for example, the word *removal* in *the removal of the last paragraph* is not a syntactic constituent.

The correctness of (16) is, however, independent of the status of the traditional orthographic word, of whether or not, in any particular case, an orthographic word does or does not correspond to a syntactic constituent. If (16) is correct, then whatever the hierarchical structure might be, in any sentence in any language, that hierarchical structure must determine the linear order.¹⁰

4. Hierarchical structure and word order variation.

Chomsky (2020, 31) has emphasized that one hears linear order, but not (hierarchical) structure. Another way to put that would be to say that for any sentence in any language the particular word order of that sentence is a given, an observable fact, in a way that a particular hierarchical (constituent) structure never is. The syntactician knows that any given sentence must be associated with some hierarchical structure; what hierarchical structure that is may, in a given case, be easy to establish or (at least initially) difficult to establish.

Word order varies. (I will continue to use the familiar and useful term 'word order' despite reservations expressed earlier about the notion 'word'.) Word order varies cross-linguistically, and it varies within a given language. A good example comes from particles of the English type, e.g. *up, down, in, out, on, off, back, away*. In English, in

⁹In the same spirit as Collins (2006), but as opposed to Chomsky (1970).

¹⁰In addition to its account of linear order, the LCA (Linear Correspondence Axiom) of Kayne (1994) yielded a derivation of the core of X-bar theory, prohibited a head H from having more than one specifier, assimilated adjoined phrases to specifiers (cf. Rizzi (1997), Cinque (1999) and Manninen (2005)), and banned rightward movement.

Abels and Neeleman (2009) agree that there is no rightward movement, but take that to be an 'axiom', whereas the LCA gives it 'theorem' status. Their weakening of antisymmetry to allow external merge of specifiers to the right is criticized by Cinque (2010, 126, note 20).

many cases involving a direct object, particles can readily appear in one of two word orders:¹¹

(17) You should pick that book up.

(18) You should pick up that book.

Given (16), of course, it must necessarily be the case that (17) and (18) differ in some way in their hierarchical structure; if they had identical hierarchical structure, they would necessarily have identical word order.

Cross-linguistically (within Germanic), (17) corresponds to the word order consistently found in Danish, while (18) corresponds to that consistently found in Swedish.¹² Again, given (16), the Danish counterpart of (17) must differ in hierarchical structure from the Swedish counterpart of (18). In effect here, and I suspect in general, intra-language word order differences, as in English (17) vs. (18), and cross-linguistic word-order differences, as in Danish vs. Swedish, are cut from exactly the same cloth.

Let us note in passing that the Danish vs. Swedish difference concerning the position of particles lends itself, given the substantial similarity between those two languages, to being called an instance of syntactic micro-variation, underlain by what would then be called a microparameter (or perhaps more than one). Moreover, it may well be that the amount and type of syntactic variation found just within mainland Scandinavian, or within the set of North Italian dialects, fractally resembles the more familiar macro-syntactic variation, if one abstracts away from the right details.¹³ It might then also be the case that all 'large' syntactic differences, as in e.g. polysynthetic vs. non-polysynthetic languages,¹⁴ or in analytic vs. non-analytic languages,¹⁵ are understandable as particular arrays built up of small differences of the sort that might distinguish one language from another very similar one. In other words, the following is plausible:¹⁶

(19) All parameters are micro-parameters.

Returning to (16), it is worth noting explicitly that although (16) says that for a given hierarchical structure only one word order is available, it does not at all say that for a given word order there is only one hierarchical structure available. To take just one example, consider the following pair:

(20) The little boy was sitting on his jacket.

(21) The little boy was putting on his jacket.

¹¹But not always:

i) Unfortunately, they had to close up shop.

ii) *Unfortunately, they had to close shop up.

with (i) vs. (ii) arguably related to Diesing (1992) on definites scrambling higher than indefinites, as well as to the discussion in Kayne (1998b, sect. 4.3) of:

iii) What he looked up was he looked up a linguistics term.

iv) ??What he looked up was he looked a linguistics term up

¹²Cf. Holmberg and Platzack (1995, 203).

¹³Cf. Poletto (2011).

¹⁴Cf. Baker (1996).

¹⁵Cf. Huang (2010).

¹⁶Cf. Kayne (2013, note 23). For a different view, see Baker (2008). For general discussion of parameters, see Biberauer (2008).

The word order looks the same, but the hierarchical structure is clearly different, as shown by:

(22) the jacket on which the little boy was sitting

(23) *the jacket on which the little boy was putting

In (20), *on his jacket* is a phrase and can be fronted; in (21), it is not a phrase and therefore cannot be.

It is worth noting that a difference in structure between (20) and (21) must hold even if *on* is exactly the same element in both sentences. Although *on* in (20) is called a preposition, and *on* in (21) is usually called a particle, Emonds (1972) explicitly takes English particles to be surface intransitive prepositions. A variant of his proposal, for the case of (21), would take *on* in (21) to be a transitive preposition, in the sense of being followed by a silent HIMSELF, as seems plausible.¹⁷ That *on* should be the same element in both (20) and (21) would in fact be expected from the anti-homophony perspective of Johns (1992, 84), Embick (2003, 156) and Kayne (2019a, 137).

5. Silent elements.

The presence of silent elements in syntax (whose presence in general facilitates the work of the semantics)¹⁸ bears on Chomsky's (2020, 31) saying that one hears linear order; this is so insofar as one by definition does not directly hear silent elements. In effect, then, one hears the linear order only of those elements in the syntax that are pronounced.

At the same time, it may be that in some cases silent elements have an effect on the pronunciation of other parts of the sentence. A striking case was given by Ronat (1977) for French. Normally, the plural -s of the French plural definite article *les* must be pronounced (as /z/) if the immediately following noun or adjective begins with a vowel, as it does in:

(24) *les adjectifs* ('the adjectives')

(25) *les autres personnes* ('the other persons')

Now *orange* in French (as in English) can look like either a noun or an adjective:

(26) *les oranges* ('the oranges')

(27) *les voitures oranges* ('the cars orange' = 'the orange cars')

Adjectives in French can lose their accompanying noun to NP-deletion (as is sometimes possible in English):

(28) *les oranges* ('the orange' = 'the orange ones')

¹⁷On HIMSELF, cf. Emonds (1972, 555) own proposal on SOMEWHERE as the object of particles. Relevant also is Kayne's (2016, 9) proposed analysis of *away* as bimorphemic, along with Collins (2007, 28) taking particles to be modifiers of a silent PLACE, in the manner of Terzi (2010). (In addition, Collins has locative particles co-occurring with a silent TO or AT.)

¹⁸The availability of silent elements has, in addition, the consequence that the need for recourse to 'direct competition' (cf. Halle and Marantz (1993; 1994) and Zompì (2019)) as a means of understanding suppletion or syncretism can be substantially reduced (perhaps to zero) - cf. Kayne (2008b; 2018a; 2019b; to appear) and Collins and Kayne (2023).

What Ronat pointed out was that although the *-s* of *les* must be pronounced in (26), with *oranges* a noun, it need not be pronounced in (28), since *oranges*, as an adjective in (28), does not, strictly speaking, immediately follow *les*, given the presence in (28) of a silent N between them (just as the pronounced N *voitures* in (27) occurs between *les* and *oranges*).¹⁹

This French case recalls the apparent blocking effect of VP-deletion on English auxiliary reduction:²⁰

(29) The students have arrived late, just as the professors have.

(30) *The students have arrived late, just as the professors 've.

with the complication that this effect, while holding strongly for finite auxiliaries, does not hold in:

(31) We should've worked harder and you should've, too.

(32) We hafta leave early and you hafta, also.

In (32), the *to* of *have to* has been reduced to *-ta*; in (31), the *have* of *should have* has been reduced to 've.²¹

Part II. Antisymmetry and movement.

6. Headedness.

Let us now again return to (16), repeated here:

(33) Linear order is entirely determined by hierarchical structure.

The view expressed in (16) contrasts sharply with a view that was widely held in the 70s and 80s,²² to the effect that there is a syntactic 'headedness' parameter that gives languages the option of realizing a complement to either side of its associated head (and similarly for specifiers). A commonly cited example of such a parameter was the case of English vs. Japanese, with English having canonical VO order (i.e. pronouncing the complement of V to its right) and Japanese having canonical OV order (pronouncing the complement of V to its left).

The historically more closely related pair English and German contrast in parallel fashion (if one abstracts away from V-2 effects):

(34) Mary has seen the book.

(35) Maria hat das Buch gesehen. ('M has the book seen')

The German direct object *das Buch* precedes the (participial) verb *gesehen*, whereas in English the direct object necessarily follows:

(36) *Mary has the book seen.

. This headedness-based way of thinking about VO vs. OV took it for granted that English-type VO sentences and Japanese- or German-type OV sentences have the same constituent structure. More specifically, a standard assumption was that in both

¹⁹If Kayne (2006) is on the right track, silent VOITURES must have moved higher; if so, the silent element allowing non-pronunciation of *-s* in (17) is actually a silent copy of a silent element.

²⁰Cf. Lobeck (1995) and references cited there.

²¹Alternatively, 've here corresponds to a reduction of *of*, as in Kayne (1997).

²²In that context, Travis (1989) was notable for 'mixed headedness'.

(34) and (35) the verb and the object are sisters, with VO and OV corresponding, in this view, to head-complement and complement-head configurations, respectively.

7. Is O really sister to V?

Further discussion is immediately called for. First, there is the basic question whether what we call objects of verbs are really externally merged as complements of V. There are clearly various cases in which what might once have been thought to be the object of a verb is better analyzed as originating as the subject of an embedded constituent, one well-known example being that of ECM sentences such as:²³

(37) Everybody believes John to be a liar.

Here, *John* is without any question best understood as being externally merged within the embedded infinitival sentence (and very likely subsequently raising into the matrix). In no sense, then, is *John* itself in an ECM-sentence like (37) the complement of *believe*. The same holds for the partially different case of:

(38) We would like that book not to be on the reading list.

Again, *that book* originates within the embedded sentence and is not, despite appearances, externally merged as the complement of *like*.

In much the same way, we can think of the small clause tradition that involves sentences like:²⁴

(39) Everybody considers/believes John a liar.

(40) We want you out of here immediately.

(41) You're making them unhappy.

in which *John*, *you* and *them* are best understood to be externally merged within the embedded small clause, rather than being taken to be true complements of the matrix verb.

Related to both the ECM and small clause cases is the proposal that certain kinds of psych-verb sentences are hidden instances of causatives.²⁵ A sentence like:

(42) The loud noise has upset the children.

would from that perspective have an analysis approximately along the lines of:

(43) the loud noise has CAUSED [the children TO BE upset]

with *upset* raising/incorporating to silent CAUSE (and with *the children* perhaps raising into the matrix part of the structure). If so, *the children* in (42) is not the complement of *upset*, despite looking like one.²⁶

In addition, double object sentences such as:

(44) Mary's talk has given Susan an idea.

could be thought of in parallel fashion with a silent CAUSE, and/or as an instance of a small clause.²⁷ In either such case, *Susan* in (44) would not itself be a complement of *give*.

²³For extensive discussion, see Postal (1974).

²⁴Cf. Williams (1975) and much subsequent work.

²⁵Cf. Chomsky (1970, (12)) and Ronat (1972), among others.

²⁶Silent CAUSE, together with a silent causer, is also found in the rather different case of:

i) The flowers are in the kitchen on purpose.

Cf. Kayne (2023, sect. 3.1).

²⁷Cf. Kayne (1981a), Larson (1988; 1990), among many others.

A partially similar analysis is also likely to be valid for sentences like:²⁸
(45) The department has provided Susan with good working conditions.
with *Susan* not being a complement of *provide*, as well as for:
(46) He put the book on the table.
as in Larson's (1988; 1990) VP-shell analysis, which explicitly has *the book* starting out in specifier, not complement, position.

The array of cases represented by (37)-(46) even makes one wonder if there are any valid cases at all in which what we call the object of a verb is truly its complement in the X-bar theory sense of the term. The only remaining cases, it seems, are those in which a verb has a single object, with nothing else in the VP, e.g.:

(47) John broke the window.

(48) The dog saw the cat.

The first of these also lends itself, though, to an analysis based on silent CAUSE. The second might involve a silent preposition.²⁹

It seems possible, then, that no verbal object, in the traditional sense of the term, is ever the true complement of that verb.³⁰ If so, then a headedness parameter formulated in terms of the relative order of head and complement can have no relevance to the question of the relative order of verb (V) and object (O), which is exactly what (33), repeated here:

(49) Linear order is entirely determined by hierarchical structure.
would lead us to expect.

8. Order differences are all due to movement.

Yet English and German do visibly differ with respect to VO vs. OV, as seen in (34) and (35), even if they share the (perhaps universally valid) property that O is not the complement of V. If we adopt Baker's (1988, 46) UTAH principle, we arrive at the further conclusion that corresponding English and German Os do not differ in their external merge position, whatever it may be.

We are left, then, if there is no external merge difference, nor any headedness parameter, with the entirely plausible conclusion that English and German must differ with respect to what movements of O and V they allow. In effect, German, like Japanese and other OV languages, must allow O to scramble to a position higher than the final position of V in a way that English does not.³¹

It should be noted, given all of the preceding, that such scrambling movements will all be Spec-to-Spec, if all Os start out in Spec positions. A potentially interesting strengthening of this conclusion would be as in:³²

²⁸For relevant discussion, see Hoekstra and Mulder (1990).

²⁹Cf. Gruber (1965).

³⁰If Kayne (2008a) is correct, nouns, on the other hand, never have complements at all; there is a basic distinction between nouns (which do not project) and verbs (which do project); other potential categories (adjectives, adpositions,...) must ultimately be either a subtype of noun, a subtype of verb, or complex.

On the fundamental status of N vs. V in signed languages, cf. Abner et al. (2019).

³¹On scrambling in Japanese, see Arano (2022) and references cited there. English may have some overt Case-licensing movement for objects, as in Johnson (1991).

³²Cf. Kayne (1998b, (136)).

(50) Complements can never be moved.
presumably for PIC-like reasons, if true.³³ Apparent counterexamples such as VP(-like) movements would then need to be analyzed as involving movement from a Spec position, in a way that would recall Haddican (2007, 751) on the requirement that focalized VPs be nominalized.

If the difference between English VO and German OV necessarily rests on differing movements of O and/or V(P), the following more general conjecture (which again rests in part on Baker's (1988, 46) UTAH principle) comes to mind:

(51) All cross-linguistic word order differences are traceable back to movement differences.

(This conjecture rests in part on the familiar assumption that we can 'match' words across languages; a word order difference comes to the fore when we have the same words, but in a different order.³⁴)

Let me note in passing that whether or not the various movements in effect referred to in (51) are 'triggered' is a partially separate question.³⁵ Also partially separate is the question of 'shape preservation' discussed by Shu (2018) and references cited there. If taken as a general principle, shape preservation (which could perhaps be formulated in a way compatible with antisymmetry) would have to come to grips with assorted apparent counterexamples.³⁶

Returning to (51), a natural extension of it would be as follows:

(52) All cross-linguistic morpheme order differences are traceable back to movement differences.

For example, English prefixal reversive *un-*, as in *undo*, *unpack*, seems to have a (partial) suffixal counterpart in Bantu languages.³⁷ From (52) it would then follow that English and the relevant Bantu languages must differ either in V(P)-movement or in the movement of this (counterpart of) *un-*, or in both.

9. Arguments must move.

Saying that English and German differ in VO vs. OV as a function of movement differences leaves open, as does (51) more generally, the question whether or not O or

³³On the PIC (phase-impenetrability condition), see Chomsky (2008).

³⁴Of additional interest is the fact that for (38) to hold, word order differences must in all probability be calculated taking into account silent elements/words; relevant here is Tang and Cheng (2014)'s idea that Cantonese 'V DO IO' has a deleted P; cf. in part Stowell (1981) and Holmberg and Platzack (1995, 219ff.), as well as Biggs (2016) on Liverpool English.

³⁵The triggering of movement was of little importance in Kayne (1994) (cf. p.140, note 3), and has been turned away from in Chomsky et al. (2019, 237-238).

³⁶Such as negative phrase movement in Scandinavian (cf. Christensen (1986) and Svenonius (2000)), stylistic fronting in Icelandic (cf. Ingason and Wood (2017)), the movement of *tout* ('all') and *rien* ('nothing') in French (cf. Kayne (1975, chap. 1), many instances of the movement of object clitics in Romance languages (cf. Kayne (1991)) and V-2 of the Germanic sort. On the partially similar Fox & Pesetsky (2005), see also Holmberg (2005).

³⁷Cf. Schadeberg (2003, 79).

V can fail to move at all in one language or another. The following conjecture is, then, worth consideration, in particular for the case of arguments that look like full DPs:³⁸

(53) Arguments invariably raise at least once, in all languages.

The question now arises as to whether or not (53) could be taken to extend even to cases in which it seems that there is an argument that is a NP, but not a DP, as in English compounds and in various types of incorporation cross-linguistically. Of relevance is the fact that there are instances of D-stranding under noun-incorporation to verbs, as discussed in Baker (1996, 299, 313, 334, 413) for Mohawk and Southern Tiwa. Baker's examples suggest to me the possibility that all instances of N(P)-incorporation involve a (stranded) D, which will often be silent. If so, then N(P)-incorporation will fit into (53).³⁹

Baker (1996, 307) points out the (partial) similarity between compounding of the English sort and incorporation of the sort just discussed. And in fact there is evidence from English compounding that leans in the direction of a silent D. I have in mind contrasts like the following:⁴⁰

(54) They just love *(the) Hudson River.

(55) They're real (*(the) Hudson River-lovers.

The fact that *the* is obligatory in (54) makes it plausible to take (55) to contain a silent counterpart THE.

The presence of a silent THE might then be extended to non-proper-name compounds like:⁴¹

(56) They're real magazine-lovers.

³⁸The argument phrases that Cardinaletti (2001) takes to be *in situ* must have been moved, if the text proposal is correct, in which case we would interpret 'in situ' in the cases she discusses as 'in their canonical position'.

The idea that argument phrases must raise at least once recalls Moro's (1997, 139) proposal that "predicates apply only to variables at LF", a proposal that took the trace/copy theory of movement to be correct, whereas my 'raise at least once' proposal is neutral as to the correctness of that theory. A second difference has to do with Kayne's (1981b) interpreting the locality difference between French object clitic movement and French *tout*-movement to reflect the fact that *tout*-movement is linked to a variable while clitic movement is not.

Perhaps obligatory raising reflects the language faculty's counterpart of a motion detector. Arguments, at least, are visible only if they move, or only as they move.

³⁹Especially if NP is in general the true argument (which it will be if the DP-hypothesis is not correct; cf. Bruening (2020), Chomsky et al. (2019, 248)).

⁴⁰Cf. the absence of a pre-prefix on Lubukusu adverbials discussed by Wasike and Diercks (2021), which might allow taking the prefixes on those adverbials to be noun class markers; cf. also:

i) That stream is very (*(the) Hudson River-like.

Note that English allows separation of noun from verb by a particle:

ii) You're an avid music downloader.

with the pre-V position of *down* in (ii) recalling the obligatory pre-V position of the particle in Swedish participial passives as discussed by Lundquist (2014).

⁴¹Relevant here is Schwarz (2013).

thinking of:

(57) The magazine has long been an important part of American culture.
which suggests that silent THE in (56) would be generic-like,⁴² as is overt *the* in (57).

Somewhat similar in having THE might be:

(58) They would like to have a third child.
against the background of:

(59) They would like to have a child that would be the third one.
with (58) then to be analyzed in the direction of:

(60) ...have a BE THE third child
in which 'BE THE third' would be a reduced relative.

In approximately the same vein, we can note:

(61) That was a most important book.
(62) That was a book of the utmost importance.
with this pair suggesting that we take (61) to be as in:⁴³

(63) ...a THE UT- most important book
If so, we would correctly not expect to have:

(64) *That was a finest book.
given the contrast between *utmost* and **ut-est*. Nor would we now expect to have, in a way parallel to (61):

(65) *That was a least important book.
given:

(66) *That was a book of the utleast importance.

10. V(P)-movement

The conjecture given in (53), to the effect that arguments must move at least once, gives natural rise to the following more tentative one:⁴⁴

(67) V(P)s invariably raise at least once, in all languages.
That verbs can raise is well-known from Pollock (1989) and Cinque (1999), among others,⁴⁵ not to mention from discussions of V-2 in Germanic.⁴⁶ On verbal participle

⁴²And/or like the definite article of French partitives, as in:

i) Nous achetons de la bière. ('we buy of the beer' = 'we're buying beer')
Cf. Kayne (1975, sect. 2.9).

⁴³There might also be a silent TO, thinking of:

i) the leftmost door
ii) the door (the) most to the left
In addition, *ut-* may be closely related to *utter*, as in:

iii) an utter failure; utterly wrong
leading to the possibility of:
iv) ...a TO THE UTT-ER most important book
with this *-er* the same as that of *outer*.

⁴⁴I leave for future work the corresponding question for adjectives and adpositions, neither of which is a core category, if Kayne (2008a) is on the right track.

⁴⁵Cf. also Williams (1974, 222). For a possible reinterpretation of Pollock (1989) in terms of phrasal movement, see Jayaseelan (2010).

⁴⁶Cf. den Besten (1983).

movement, see Bošković (1995), as well as Cinque's (1999, Appendix 1, A2) discussion of multiple variations on that theme in Italian dialects. On the comparative syntax of infinitive-raising within Romance and its effects on PRO, see Kayne (1991).⁴⁷ Moreover, Johnson (1991, 607) has argued that verb-raising is found even in English. As for VP- or vP- movement, see the various papers in Lee-Schoenfeld and Ott (2021), as well as J. Collins (2017).

One would of course like to know why the language faculty would be so fond of verb movement, in particular if (67) were to turn out to be correct. A potential answer, one that would be linked to (53), might bring into play Holmberg's generalization (and Johnson's (1991, 607) extension of it to English),⁴⁸ to the effect that verb movement is a necessary precondition for object shift/movement. A universalizing reinterpretation of Holmberg's generalization would, given (53), force the existence of verb movement universally, perhaps especially, thinking of the discussion of (37)-(46), if all verbal objects without exception start out as subjects of constituents lower than V.⁴⁹

Verb movement itself is an instance of head movement, the validity of which has been questioned,⁵⁰ in a way that rests of the idea that head movement must be an instance of adjunction. Were head movement not adjunction, that objection to it would fall away.⁵¹

11. OXV

I suggested earlier that German OV sentences, and OV sentences in general, have the O in a Spec position, rather than in the complement position of V. That O is in a Spec position is particularly clear in OV languages in which O canonically precedes V, yet without immediately preceding it.⁵² What I have in mind here are languages that have, in whole or in part, canonical/neutral OXV order, for some pronounced X.

One type of OXV language has X = Neg; in such languages ONegV is a possible canonical order (as in Korean). As noted by Whitman (2005) for both Korean and Japanese, it must be the case, on the standard assumption that Neg is externally merged outside VP (and therefore above the external merge position of O), that the pre-Neg position of O in SONegV sentences is the result of the movement of O to the Spec of some head higher than Neg. If so, then in a SONegV sentence, O is clearly not occupying the complement position of the pronounced V.

Whitman (2005) argues more specifically that SONegV is produced by remnant VP-movement.⁵³ The verb moves out of the VP by head movement; subsequently the

⁴⁷On the micro- side, cf. Kayne (1991, (text to) note 69; 1996).

⁴⁸See Holmberg (1986; 1999) (and Nilsen's (2003, chap. 3) alternative), as well as Déprez (1994, 135).

⁴⁹Note that the unaccusative hypothesis, especially going back to Perlmutter (1978, sect. 4; 1989) and Burzio (1986), combined with the Hale and Keyser (1993; 2002) view that unergatives invariably derive from transitives, leads to the conclusion that all verbs have objects.

⁵⁰On head movement, see Mahajan (2000), Matushansky (2006) and Chomsky et al. (2019, 249) and references cited in those works.

⁵¹For possibly relevant discussion, see Kayne (2016).

⁵²These appear not to have been taken into account by Liao (2023) or Zeijlstra (2023).

⁵³Cf. also Koopman (2005b).

entire (now verbless) VP containing O moves past Neg, much as in Nkemnji's (1992; 1995) analysis of one word order pattern in Nweh.

An alternative to remnant VP-movement for SOXV is to have O move past X by itself. Kandybowicz and Baker (2003) in fact argue that both options are made available by the language faculty. While remnant VP-movement is appropriate for Nweh and also for Lokaa, movement of O by itself is called for in Nupe, with this difference reflected in the fact that Nweh and Lokaa allow S-PP-X-V, whereas Nupe does not.

Lokaa also allows sentences with OAuxV order. This order is another clear instance in which O cannot possibly be in the complement position of V. Such OAuxV sentences are also found in limited cases in Dutch and German, in particular in (embedded instances of) so-called IPP ('infinitive-for-participle') sentences, as discussed by Zwart (2007) and Abels (2016) among others. An IPP example from German is:

(68) Ich glaube, dass er das Buch hätte lesen wollen. ('I believe that he the book would-have to-read to-want' = 'I believe that he would have wanted to read the book') In this example, the object *das Buch*, which is the thematic object of *lesen* ('to-read'), ends up immediately preceding, not *lesen* itself, but rather the auxiliary *hätte* ('would-have'), which then in turn precedes the lexical verb *lesen*. (The reason for the IPP label lies in the fact that *wollen* ('to-want') in (68) is in infinitival form, rather than in the participle form that usually cooccurs with an auxiliary like *hätte*.)

Another subtype of canonical OXV, in addition to the cases where X = Neg or Aux, is OFocV, found in languages in which O normally precedes a VP-external focussed phrase (if there is one) that itself precedes V. This subtype has been argued by Jayaseelan (2001) to match that of Malayalam, in which objects must surface in a position preceding that of VP-external focus.

Somewhat differently, returning to Dutch and German, we can note that in both those languages the infinitive marker (*zu* in the German example below) must intervene between object and following verb and so, too, must (abstracting away from V-2) what are called separable particles (*mit* in the following example), with an example being:

(69) ...das Buch mitzubringen. ('...the book with to bring' = '...to bring along the book') The object *das Buch* here precedes the infinitival verb *bringen*, but does not immediately precede it, and is clearly not in a sister position to it. It is separated from its verb in this canonical order by both the infinitive marker *zu* and the German separable particle *mit*. Somewhat similarly, in another Germanic language/dialect, West Flemish, objects in embedded sentences must precede and be separated from the following verb by one of the negation markers that West Flemish has, as discussed by Haegeman (2001; 2002).

The various languages mentioned in this section thus add support to (53) and to the idea that O, even if it were to start out in a complement position, could never end up in one.

Part III. Evidence for antisymmetry.

Section 1. General points.

12. Left-right asymmetries.

Let us now return to (51), repeated here:

(70) All cross-linguistic word order differences are traceable back to movement differences.

In the specific case of OV vs. VO, this just means that some languages will have a combination of movements such that O ends up higher than V, while in others an at least partially different combination of movements will have V ending up higher than O. There is no need for a directionality parameter.⁵⁴

The example of OV/VO cross-linguistic word order variation, though a convenient starting point, should not, however, be allowed to give the impression that studying word order variation pairwise is sufficient. That it is not sufficient was in effect demonstrated by Greenberg (1966) in his study of DP-internal word order.

Greenberg found that if you look, for example, at the pair demonstrative (Dem) and noun (N), you find that both ordering possibilities (Dem N, as well as N Dem) are attested cross-linguistically. And the same holds for numeral (Num) and N, with Num N order attested in some languages, and N Num order in others. Similarly for the relative order of adjective (Adj) and noun, insofar as both Adj N and N Adj orders are attested. Looking at these three pairs separately gives the initially misleading impression that the language faculty is word-order-wise extremely flexible.

Yet, as Greenberg showed in his Universal 20, if you instead examine combinations in which all four of these categories occur together, the picture changes considerably. An asymmetry suddenly appears, as follows. If N is final, then the order of the other three is fixed, and one has only 'Dem Num Adj N' order. Whereas if N is initial the order of the other three is not fixed.

As Cinque shows in papers and in a monograph,⁵⁵ this asymmetry between prenominal order and postnominal order fits directly into the antisymmetry proposal that I made in Kayne (1994), from the perspective of which such left-right (pre-N/post-N) asymmetries are expected. As Cinque (2009; 2023) further shows, such complex left-right (pre-X/post-X) asymmetries are in fact more widely found than just in the case corresponding to Greenberg's Universal 20. And as in Greenberg's case, these asymmetries come to the fore only if one examines the relative order of larger sets, not just pairs, of elements.

13. Antisymmetry as a generalization over Greenbergian generalizations.

The proposal in (70) does not by itself tell us how hierarchical structure (whether produced by internal merge or by external merge) fits together with linear order. As noted earlier in section 2, antisymmetry does provide a specific answer, giving the relation as in:

(71) S [H C]

i.e. for any phrase with head H, the specifier S must precede H and the complement C must follow H. (In addition, antisymmetry has it that a head H can have only one specifier and adjoined phrases are assimilable to specifiers.)

Antisymmetry, which can be seen as a kind of anti-optionality, insofar as it says that a given hierarchical structure allows for no options as to how it maps to linear order, is supported by a variety of empirical arguments that often involve cross-linguistic gaps,

⁵⁴Cf. Kayne (2011).

⁵⁵For details and theoretical underpinning, see Cinque (2005; 2020a; 2023).

i.e. a priori plausible word order patterns that in fact are not found. (In the study of syntax, as we know, it is essential to see and examine what is not there). Antisymmetry can thus be thought of as grouping together a substantial set of Greenbergian cross-linguistic generalizations and providing a single theoretical account for all of them.

In the following sections, I will discuss a number of such cross-linguistic generalizations.

Section 2. Phrases in high(ish) positions.

14. Hanging topics.

As an initial example, let me take Cinque's (1977) demonstration that Italian has two distinct types of left-dislocation, one of which he calls "hanging topics." Hanging topics occur at the very left-hand edge of a (non-embedded) sentence. As far as I know, there has never been a claim to the effect that there exists something exactly comparable on the right-hand edge of the sentence, in any language.

If we now ask why there should be left-hand (sentence-initial) hanging topics, but no right-hand (sentence-final) hanging topics, we can easily see, from an antisymmetric perspective, that the core reason is the general prohibition against right-hand specifiers:

(72) S [H C]

vs.:

(73) *[H C] S

A hanging topic in a maximally high Spec position can be sentence-initial, given (72), but cannot be sentence-final, in light of (73).

Cinque's Italian examples involve hanging topics that are DPs. A non-DP (partial) counterpart from English is given in:

(74) As for John, we like him a lot.

(75) *We like him a lot, as for John.

An English *as for*-phrase appears in a high Spec position, necessarily to the left, rather than to the right.⁵⁶

As Cinque (1977, note 4) notes, hanging topics in German show no Case agreement, as seen in:

(76) Der Professor, Sie lobten ihn. ('the professor (nom.), they praised him (acc.))' and similarly with a dative pronoun (example from Josef Bayer (p.c.)):

(77) Der Hans, Sie hat ihm ein Buch gegeben. ('the Hans (nom.), she has him (dat.) a book given')

In both of these, the hanging topics (*der Professor*, *der Hans*) have nominative Case, even though the associated pronouns (*ihn*, *ihm*) have accusative and dative Case, respectively. As Bayer notes, a right-hand hanging topic counterpart of these examples with non-agreeing Case would be "totally out", e.g.:

(78) *Sie hat ihm ein Buch gegeben, der Hans.

15. CLLD and CLRD

⁵⁶Note that in such cases leftward movement past the *as for*-phrase is not available to English (nor, possibly, to any other language) for reasons that will need to be made precise.

An if anything even more striking example of a left-right difference that jibes with antisymmetry comes from clitic left-dislocation (CLLD) of the sort discussed in detail in Cinque (1977; 1990) for Italian, in which a left-dislocated phrase is linked to a following pronoun, somewhat as in the following English example.⁵⁷

(79) That guy, he's real smart.

Italian CLLD does at first glance seem to have a right-hand counterpart, called (clitic) right-dislocation (CLRD), much as in this English example:

(80) He's real smart, that guy.

Yet any symmetric pairing of CLLD and CLRD in Italian would be misleading. As argued in detail by Cecchetto (1999) for Italian and similarly by Villalba (1999) for Catalan,⁵⁸ there are sharp asymmetries within each of those two languages between CLLD and CLRD, a surprising fact unless the linguistic universe is antisymmetric.

For example, Cecchetto shows that the CLLD phrase (the Italian CLLD counterpart of *that guy* in (79)) asymmetrically c-commands subject position, while the CLRD phrase (the Italian counterpart of *that guy* in (80)) is asymmetrically c-commanded by subject position. This difference in hierarchical structure in Italian feeds into Condition C differences among others.⁵⁹ From the perspective of antisymmetry, this reflects the basic S-H-C configuration that allows the CLLD phrase to occupy a high Spec position in the left periphery, in Rizzi's (1997) sense. Whereas the CLRD phrase has, by antisymmetry, no parallel right-hand Spec position available to it.

More specifically, the antisymmetric prohibition against right-hand specifiers forces on CLRD either a remnant movement analysis of the sort that Cecchetto proposes for Italian, with the CLRD phrase moving to a left-hand VP-peripheral topic position,⁶⁰ or else a bi-clausal analysis of the sort that I had suggested for English in Kayne (1994, sect. 7.3), with (80) then related to:⁶¹

(81) He's real smart, that guy is.

which in turn may well contain a silent SO, thinking of the existence of:

(82) He's real smart, so he is.

Finally, we can note that, unsurprisingly now, the CLRD-related (81) has no CLLD counterpart:

(83) *That guy is, he's real smart.

16. Heavy-NP-Shift

Worth noting explicitly is how CLRD as in (80) illustrates the point that the systematic left-hand position of specifiers relative to their own head (S-H-C) in no way precludes a phrase in Spec (*that guy*, in (80)) from ending up at the end of a sentence. Although every specifier position must be to the left of its associated head position, the

⁵⁷On the only in part similar Italian 'emarginazione', see Cardinaletti (2001).

⁵⁸Cf. also Villalba (2007).

⁵⁹Cecchetto mentions Benincà's (1988) showing that there is also a subtle semantic difference between the topic in CLLD and the one in CLRD.

⁶⁰Cf. also Belletti (2004).

⁶¹Cf. Tanaka (2001) on right-dislocation in Japanese, Ott (2014) on German and Lee (2023) on Alasha Mongolian; in defense of a remnant movement approach, cf. Chiang (2022) on Mandarin.

relative order of the elements in those positions can be disrupted by movement (internal merge); in particular the head may move past its associated specifier either by head movement or, as in Cecchetto's remnant movement analysis, by phrasal movement.

Alternatively, under a biclausal analysis that would have (80) derived from (81)/(82), *that guy*, though in a left-hand Spec, ends up appearing at the right-hand end of the (visible part of the) sentence as the result of deletion,⁶² with (80) then having an analysis as in:

(84) he's real smart, that guy IS REAL SMART

Other cases of phrases ending up at the right as the result of leftward movement across it are arguably not hard to find. An early notable example is Larson's (1988; 1990) reanalysis of Heavy-NP Shift as not involving rightward movement,⁶³ but rather as involving leftward movement across the 'heavy' phrase. The derivation of a sentence like:

(85) They put on the table the book that had fallen to the floor.
has *put on the table* moving leftward across *the book that had fallen to the floor*.

Subsequently, den Dikken (1995) proposed a revision of Larson's analysis such that the derivation of (85) must contain an additional scrambling step that applies to *the book that had fallen to the floor* prior to the preposing of *put on the table*. The existence of such an additional scrambling step is supported by the following preposition-stranding facts:

(86) ??I should be able to look at in a day or two the article you just sent me this morning.

(87) *John will introduce you to in a day or two the woman he's planning to marry.

In (86), *at* is stranded by the scrambling of *the article you just sent me this morning*; the result is marginal.⁶⁴ In (87), *to* has been stranded by the scrambling of *the woman he's planning to marry*; the result is strongly deviant. The difference in acceptability status between (86) and (87) is plausibly due to the presence of the direct object *you* in (87), which arguably acts as an intervener for the scrambling in question. Insofar as the pre-Larsonian rightward-movement approach to Heavy-NP-Shift could not account for (86) vs. (87) at all, we have a clear argument in favor of the Larson/den Dikken leftward-movement-based account.

In the derivation of (85), the preposing of *put on the table* finds a landing site somewhere below subject position. Also possible is a landing site above subject position, in sentences like:

(88) ...and put on the table they did the book that had fallen to the floor.
That 'heaviness' is at issue is shown by the contrast with:

(89) ...and put on the table they did ??that book/*it.

Again, in (88) the object phrase *the book that had fallen to the floor* ends up sentence-final, in a way perfectly compatible with its being in a left-hand Spec position.

⁶²Deletion may itself involve movement - cf. Johnson (2001) and Kayne (2022a).

⁶³For early discussion of heaviness effects, see Fraser (1976) and Kroch (1979). That they might have a syntactic (as opposed to processing) basis was suggested in Kayne (1998a).

⁶⁴But to my ear not as deviant as Bresnan (1976, 33) takes it to be.

Another important property of ‘Heavy-NP-Shift’ sentences like (85) is that some languages don’t allow them. This may not seem surprising when it comes to strongly OV languages like Japanese, as discussed by Saito and Fukui (1998). But things are a bit more complicated. German and Dutch are strongly OV, too (abstracting away from V-2), and in fact it has been said by Groos and van Riemsdijk (1981, 184) and by Hirschbuhler and Rivero (1983, 515) that German and Dutch by and large lack Heavy-NP-Shift. On the other hand, Hagen Blix (p.c.) has told me that German allows:

(90) Gestern hat den Hund ausgeführt der Mann, der... (‘yesterday has the dog taken-out-for-a-walk the man who...’)

in particular if the post-verbal DP (*der Mann der...*) is contrastive relative to expectation.

If German allows (90) in a way that Japanese doesn’t, that would suggest a link to the fact that German, even in non-V-2 contexts, also readily allows postverbal sentential complements, postverbal comparative clauses, postverbal ‘extraposed’ relative clauses and postverbal adjunct clauses, all in a way that Japanese doesn’t. This Japanese vs. German difference is in turn arguably related to the fact that German has initial complementizers and initial adjunct-introducing elements comparable to English *since*, *before* et al. If so, then these German vs. Japanese differences are not irreducible ones.

‘Heavy-NP-Shift’ is, however, also absent from various VO languages, such as Haitian Creole, Bambara, Chinese and Gungbe, according to Dejean (1993), Koopman (1992, 581), Tang (1998, 132) and Enoch Aboh (p.c.), respectively, for reasons that may have to do with the DP-final position of D.⁶⁵

17. The (un)masking of leftward movement.

Heavy-NP-Shift sentences illustrate instances of phrases ending up toward the right -hand edge of the visible part of the sentence as the result of leftward movement across them, without any rightward movement being involved. In this section, I will briefly mention a number of other such instances. One can be seen in Buell’s (2011) discussion of a Zulu counterpart *ngani* of English *why*, which he demonstrates is in the Comp area in Rizzi’s (1997) sense, and yet at the same time is in postverbal position, as the result of phrasal movement of IP to the specifier of Int⁰ (p. 821).⁶⁶

A second such case is discussed by Poletto and Pollock (2004a; 2004b) (and references cited there) for the NorthEast Italian dialect Bellunese, which has interrogative sentences such as:

(91) Ha-tu magna che? (‘have you eaten what’)

Poletto and Pollock show in detail that despite initial appearances the sentence-final wh-phrase *che* in (91) has been raised leftward into the Comp area. Its postverbal position in (91) is due to remnant IP-movement having subsequently raised (at least) the verb *magna* past it. Worth noting, in addition, is the fact that Bellunese also allows interrogative doubling sentences:

⁶⁵For discussion relevant to this and to German vs. Japanese, see Kayne (1998a; 2003).

⁶⁶A challenge for future work is to understand why *ngani* can be postverbal, while subject and object agreement markers/clitics are always preverbal in (narrow) Bantu languages.

(92) *Cossa ha-lo fat che?* ('what has he done what')
which Poletto and Pollock argue to instantiate movement to two distinct positions in the left periphery.

As a third example of movement to a high(ish) Spec position masked by subsequent V(P)/IP-movement, we can take Belletti's (2004) analysis of Italian sentences like:⁶⁷

(93) *Ha parlato Gianni.* ('has spoken J')
that contain a postverbal focussed phrase that she shows to be located in a focus position in the low IP area above VP.

In a way that is partially similar to the preceding three examples, Reglero (2003, sect. 5.2) shows that in Basque multiple interrogation sentences the second *wh*-phrase, although it looks as if it's in situ, is actually in a structurally higher position than would be its non-*wh* counterpart.

Somewhat similarly, Shlonsky (in press) reanalyzes what has been called subject-object reversal in Bantu languages in terms of leftward movement of the subject to a (relatively) low focus projection, followed by a 'smuggling'-type movement to a position above FocusP.⁶⁸

All the studies touched on in this section involve either *wh*-phrases or focussed phrases that appear postverbally despite having moved into a (relatively) high left-peripheral position.⁶⁹ The following section will briefly make a similar point concerning postverbal subjects that are not (necessarily) focussed.

18. Very high post-V topics and subjects.

One case concerns a DP that is more topic-like than subject-like, in Malagasy, as discussed by Pearson (2018), who argues that the predicate comes to precede that topic-like constituent (the 'trigger', in his terms) via predicate fronting.⁷⁰ Another, from French, has to do with what look like 'ordinary' postverbal subjects, as in Kayne and Pollock's (2001) discussion of French 'stylistic inversion', illustrated in:

(94) *Qu'a dit Jean?* ('what has said J')
in which the subject appears postverbally (in this example, post-participially). Kayne and Pollock (2001) argue that in sentences like (94) the subject *Jean* has raised to a

⁶⁷Cf. in part Ordóñez (1998) on Spanish VOS sentences.

An interesting twist on what ends up as post-V focus is pointed out by Watters (2003): "Babungu (Ring) (Schaub 1985:10) uses a similar strategy to that of Nweh, except that when the subject noun phrase is questioned, the interrogative word occurs following the verb and a reduplicated form of the verb follows. In this way, an SV order becomes VSV."

⁶⁸On smuggling movements, see the various papers in Belletti and Collins (2021).

⁶⁹It is not clear that such cases will be easy to reconcile with the (tentative) suggestion made in Chomsky (to appear) to the effect that "pragmatic reasons" might be involved.

⁷⁰See also the various paper in Lee-Schonfeld and Ott (2021).

Predicate raising of the sort that played a role in Kayne (1998b, sect. 1.3), going back to Koster (1994) and Zwart (1994, 399ff.; 1997, 43, 100ff.), may be relevant to the prohibition against resultative small clauses as objects of prepositions mentioned in Kayne (1985, 123), and may be in competition with Stowell (1991).

high position above IP, with the visible word order then produced by remnant IP-movement past the landing site of the subject.

The French example in (94) is then another case, as in all the paragraphs going back to (76), in which a phrase that ends up in a left-hand Spec position can have its leftness obscured by subsequent V(P) or IP movement past it, in a way consonant with the antisymmetry hypothesis of consistent S-H-C order.

19. Relative clauses.

The antisymmetric character of the language faculty can be detected in ways that go well beyond what we have seen so far. Particularly striking in this regard is the absence of symmetry between prenominal and postnominal relative clauses. One might have expected prenominal and postnominal relatives to be essentially similar, merely differing in their order with respect to the “head” of the relative. However, substantial differences in syntactic behavior between pre- and post-nominal relatives have been noted by Downing (1978) and Keenan (1985). (Let us set aside correlatives, which are not visibly headed in the relevant sense, and let us in addition confine our attention to relative clauses that occupy their canonical position in the language in question.) Then we have the following salient differences, as observed by Downing (1978) and by Keenan (1985):

(95) Prenominal relatives (as opposed to postnominal relatives) generally lack complementizers akin to English *that*.

(96) Prenominal relatives (as opposed to postnominal relatives) usually lack relative pronouns.

(97) Prenominal relatives (as opposed to postnominal relatives) tend to be non-finite.

These differences fed into the proposal that I made in Kayne (1994, chapter 8) to the effect that prenominal relatives originate postnominally. Subsequently, a particularly striking piece of evidence in favor of that proposal was discovered by Kornfilt (2000; 2005).⁷¹ She noticed that the Turkic languages Sakha and Uigur have prenominal relatives whose subjects trigger agreement such that the agreement morpheme in question actually appears following the “head” noun, despite the fact that the relative itself precedes that noun. Kornfilt’s proposal is that this unexpected agreement pattern reflects the preposing of the originally postnominal relative.⁷² There is a high subject agreement morpheme that originates within the postnominal relative.⁷³ When that relative is moved to prenominal position, Sakha and Uigur do not move the entire relative; most of it is moved, but the agreement morpheme in question is

⁷¹A similarly striking argument is given by Aldridge (2003) based on PP-stranding in Tagalog, in a way that recalls in part den Besten and Webelhuth (1990). (For an antisymmetry-compatible approach to relative clauses that does not have relatives originating postnominally, see Cinque (2020b).)

⁷²Cf. Simpson (2001, sect. 4) on the preposing of relative clauses past *de* in Mandarin Chinese.

⁷³For an at least partially different view (on Kazakh), see Ótött-Kovács (2023). How close this phenomenon is to the agreement associated with the copula in Turkish as proposed by Kornfilt (1996) remains to be determined.

stranded in a position following the head of the relative. In effect Sakha and Uigur have left us a visible clue to the original position of the relative clause.

In an antisymmetric syntactic universe, the following should turn out to be correct (as seems to be the case):

(98) No postnominal relatives ever have their subject determining agreement such that the agreement morpheme precedes the “head” noun.

The fact that Sakha and Uigur prepose less than a full relative clause is, I think, closely related to (95)-(97), which seem to be telling us that a maximally large relative clause cannot in fact be preposed from postnominal position. A preposed relative cannot contain very high elements corresponding to complementizers or to relative pronouns; nor, usually, can such preposing carry along whatever exactly it is that is realized as finiteness.

As for a more precise account of (98), part of it is that the relative clause preposing that Kornfilt (2000; 2005) uses can, given antisymmetry, have no mirror-image (postposing) counterpart. But one might still perhaps wonder about the possibility of generating, without any movement of the relative, AGR N REL, and then having AGR pick up the phi-features of the subject within REL. This is arguably impossible, however; if downward agreement exists, it could be excluded by a PIC-like effect, with the subject of REL too deeply embedded to be seen by AGR; if downward agreement without associated movement does not exist, the account is even simpler.⁷⁴

Section 3. Auxiliaries, verbs and pronominal clitics.

20. Aux and V.

Shifting gears, but only in part, we can note that Basque generally has VAux, but in the presence of (primarily matrix) negation has NegAuxXV.⁷⁵ Let me adopt the suggestion made in Cinque (1999, 189, note 20 and associated text) to the effect that Basque VAux is derived by leftward phrasal movement, starting from a structure in which Aux is above VP,⁷⁶ and that this leftward phrasal movement past Aux is blocked by a higher (primarily matrix) Neg, yielding NegAuxXV. Against this background, we can envision the following conjecture, which is much like (98):

(99) There is no language that is AuxV in general, but VXAuxNeg in the presence of negation.

In other words, there is no ‘reverse’ Basque in the relevant respect.

As with (98), part of the account of (99) rests on the antisymmetric exclusion of the possibility of any rightward movement counterpart of the leftward movement postulated by Cinque. Another part would involve the question of where exactly Neg can be externally (or internally) merged, but that would take us too far afield.⁷⁷

21. V-2 and CL-2

⁷⁴For discussion, see Kayne (2018b, (text to) note 30). On the possible non-existence of movement-less Agree, cf. also Koopman (2003; 2005a).

⁷⁵Cf. Laka (1993, 247, 263).

⁷⁶Cf. Artiagoitia (1994, 370).

⁷⁷For relevant discussion based on Romance languages/dialects, see Zanuttini (1997).

The preceding discussion of (98) and (99) ties in both with verb-second (V-2) and clitic-second (CL-2) phenomena. V-2 is well-known from Germanic languages.⁷⁸ It involves the leftward raising in (primarily) root sentences of a finite verb to a position such that the finite verb can be preceded by only one phrase. The following conjecture seems likely to hold:

(100) There is no reverse V-2.

That is, there is no language in which, in root sentences, the finite verb would be raised rightward to a position such that it could be followed by only one phrase.⁷⁹

CL-2, whose partial similarity to V-2 is emphasized by Progovac (1999a), arguably mimics (100), as in:

(101) There is no reverse CL-2.

In CL-2 languages, clitics occur near the left edge of the sentence, preceded by a single phrase (or perhaps word). What (101) says is that there is no language whose clitics must occur near the right edge of the sentence, followed by a single phrase. Put another way, CL-2 has clitics second-from-the-beginning; there is no second-from-the-end counterpart of that.

Let us note in passing that even though the apparent positioning of pronominal clitics toward the right in Irish does not directly bear on (101), insofar as there is no ‘second-from-last’ effect in Irish, the Irish phenomenon does bear on antisymmetry in a different way, since the rightward movement/postposing analysis of Bennett et al. (2016) is incompatible with the antisymmetric prohibition against rightward movement;⁸⁰ their proposal is argued against by Thoms (2021).⁸¹

22. The decomposition of CL-2 and of V-2.

Discussions of CL-2 languages, e.g. that in Bošković (2016), generally omit European Portuguese (EP). Understandably so, since EP allows pronominal clitics to be preceded by more than one phrase/element, in a way that makes EP look quite different from, say, the CL-2 Slavic languages. The following EP examples are from Kato et al. (2023, sect. 5.6.1)):

(102) Ele não me agradeceu. (‘he neg me thanked’)

(103) Talvez a Maria o contrate. (‘maybe the Maria him hires’)

(104) Ontem sempre nos rimos um bocado. (‘yesterday always us(reflex.) laughed a bit’)

And yet, EP resembles CL-2 languages in systematically prohibiting its clitics from appearing sentence-initially, apart from some (not uniformly accepted) exceptional cases noted by Kato et al. (2023, 306, note 100). Possibly, then, CL-2 is not a single

⁷⁸Going back in particular to den Besten (1977).

⁷⁹For additional discussion, see Kayne (1994, sect. 5.4).

⁸⁰Strictly speaking their proposal involves postposing as a post-syntactic phonological process, so one could imagine a post-syntactic exemption from antisymmetry; that kind of theoretical weakening would not seem desirable in general, though, and would not be compatible with Collins and Kayne (2023).

⁸¹Cf. the argument in Kayne (2000b) against McCloskey (1999), concerning a different area of Irish syntax.

property. Instead, it could be thought of as the intersection of two distinct properties, as follows:

(105) prohibition of clitics from sentence-initial position

(106) prohibition against clitics being separated from sentence-initial position by more than one phrase

EP would respect (105), but not (106). The CL-2 languages discussed by Bošković (2016) would respect both.

Transposition of (105)/(106) to V-2, thinking of Progovac (1999a) on the parallelism between CL-2 and V-2, would break classical V-2 into (at least) two parts:⁸²

(107) prohibition of finite verbs from sentence-initial position

(108) prohibition against finite verbs being separated (in root sentences, in Germanic) from sentence-initial position by more than one phrase⁸³

We are now in a position to ask what exactly we mean by ‘sentence-initial position’. German V-2 has the property, discussed in detail recently by Trutkowski (2016), that it is compatible with ‘topic-drop’, whereby a preposed topic that has induced V-2 can remain silent/be silenced under certain conditions. Trutkowski’s initial example (p. 1) is:

(109) *Wo ist dein Ring? Hab ich verkauft.* (‘where is your ring. Have I sold’)

The second half of (109) shows V-2, with the verb/aux *hab* preceded by a silent topic (whose antecedent is *dein Ring*). In English, a comparable silent topic would not be possible:

(110) *Where is your ring? *I sold.*

with a possible generalization being that such silent topics in Germanic languages are allowed only in the presence of V-2, i.e. the silent preposed (pronominal) topic in (109) is licensed by the raised finite V.⁸⁴

What this means is that ‘sentence-initial position’ in (107) is not to be calculated simply in terms of pronounced elements. The verb *hab* in (109) evidently does not count as sentence-initial, due to the presence of the silent topic preceding it.

Consequently (109) is compatible with (107).⁸⁵

Of interest now is the fact that something similar holds for CL-2, at least in Czech. Toman (1996, 505), as part of an extended argument to the effect that CL-2 in Czech is not at bottom phonological, notes that “in informal colloquial Czech, the first constituent of the clause can under certain conditions be truncated, leaving the clitic in the clause-initial position”; he then gives the examples:

(111) *To bych netvrdil.* (‘it would-I not-claim’)

(112) *Bych netvrdil.*

⁸²Basque is subject to the first of these (cf. Artiagoitia (1994, 374)), but not in any general way to the second, even in root sentences - cf. Artiagoitia’s p. 371 examples:

i) *Ordu honetan Ainhoa nora doa?* (‘time this Ainhoa where goes’)

ii) *Ez omen dator Ainhoa etxera.* (‘neg apparently comes Ainhoa home’)

⁸³Though note Nilsen (2003, chap. 3) on certain exceptions in Norwegian; also dislocations of the sort seen in examples (76) and (77).

⁸⁴Cf. Rizzi (2006).

⁸⁵As are German yes-no questions, if they contain some silent element preceding the finite verb.

in which auxiliary *bych* is the clitic that is compatible with topic-like *to* being in (112) deleted/silent. (Toman p. 506 explicitly notes the similarity to German V-2.⁸⁶) So again, as with V-2, ‘sentence-initial position’ in (105) must, in Czech, be calculated taking into account silent elements.

Returning to EP, Kato et al. (2023, 311) note that in contexts of enclisis in which a clitic would be expected to immediately follow a finite verb, there is an exception when the finite verb is indicative future or indicative conditional, in which case one gets what is traditionally called mesocclisis, as illustrated in:

(113) Eu escrever-te-ei no mês que vem. (‘I write-you-*ei* in-the month that comes’)
In (113), the pronominal clitic *te* appears between the main (infinitival) verb *escrever* and the future ending *ei*. Kato et al.’s (2016, 312) analysis, which seems right, is that *te* is enclitic to the main verb and that this *ei* is an affixal (finite) auxiliary. In effect, EP, though in general an AuxV language, has a very limited V-CL-Aux subpart, with Aux suffixal. In the spirit of (100) and (101), which took/conjectured there to be no reverse V-2 or CL-2, we can now entertain the following conjecture prohibiting the existence of a reverse EP:

(114) No VAux language will have an Aux-CL-V subpart, with Aux prefixal.

The derivation of (113) very likely will involve leftward remnant movement of (*eu*) *escrever-te* past auxiliary *ei*. The validity of (114) will then be linked to the impossibility, given antisymmetry, of any rightward remnant movement of CL-V past Aux.

23. AuxVM vs. *MVAux

Germanic languages show a striking asymmetry when it comes to AuxV vs. VAux.⁸⁷ The VAux (abstracting away from V-2, as usual) languages such as German, Dutch and related dialects often allow AuxV in special cases, as in (68) above, repeated here:

(115) Ich glaube, dass er das Buch hätte lesen wollen. (‘I believe that he the book would-have to-read to-want’ = ‘I believe that he would have wanted to read the book’)
Using M for Modal, we can think of (115) as being an instance in German of AuxVM (*hätte lesen wollen*). What is striking is that the Germanic AuxV languages (English and Scandinavian) never have special cases that would be the reverse of AuxVM, i.e. we have:

(116) Germanic AuxV languages never have MVAux as a special case.

Generalizing (116) leads to the conjecture:

(117) No AuxV language has MVAux as a special case.

As for why (116)/(117) might hold, a first approximation would be that AuxVM in a VAux language comes about via leftward V(P)-type movements that could not be

⁸⁶German(ic) V-2 is broadly limited to root sentences; the extent to which CL-2 might in some languages be limited in that way needs to be determined.

⁸⁷Other apparent asymmetries within Germanic (that I won’t attempt to go into here) include:

- i) No Germanic OV-language lacks Agr. (from Holmberg and Platzack (1995, 73))
- ii) Topicalization of negation (as with Swedish *inte*) is limited to Germanic VO languages.
- iii) Complementizer agreement is limited to Germanic OV languages. (from Koopman (2005b, note 25).

mimicked in any mirror-image fashion in an AuxV language, since rightward movements (of any type) are excluded by antisymmetry. (The details will, unsurprisingly, be more complex, as a reading of Abels (2016), van Craenenbroeck et al. (2019) and Cinque (2023, sect. 4.4) quickly confirms.⁸⁸)

24. Definite articles and clitic-second.

Let me briefly end this digression into the details of CL-2 (and away from the main point concerning the absence of any reverse CL-2, as stated in (101)), by mentioning Bošković's (2016, 25-27) proposal that CL-2 is limited to languages that lack a definite article. The first thing to notice, given the breakdown of CL-2 into (105) + (106), is that Bošković's proposed limitation, to the extent that it holds,⁸⁹ could perhaps be reinterpreted as applying to (106), but not, given EP, to (105), insofar as EP has definite articles.

The second point to be made is that the notion 'lack a definite article' itself calls for clarification, with one subpoint involving the familiar choice between 'completely lacks' and the arguably more plausible 'lacks a pronounced definite article', and a second subpoint being that the question of the existence of silent definite articles is orthogonal to the question whether articles project.⁹⁰

Section 4. Further asymmetries.

25. Word order and *have*.

A rather different sort of antisymmetry 'footprint' from any of the above is detected by Mahajan (1994). It has to do with the well-known fact that many languages lack a verb that corresponds directly to English *have*. Although the basic fact is well-known, Mahajan noticed that the presence of a verb like *have* is cross-linguistically sensitive to the (canonical) word order of the language in question. Mahajan specifically argued that *have* is found more robustly in SVO languages than in SOV languages for principled reasons.

Although Mahajan's exact proposal didn't directly allow for Dutch or German or Basque having *have*, his generalization can be tweaked and arguably improved (while maintaining the same spirit) if we say that *have* is excluded from OV languages like Hindi if they in addition entirely lack any instances of AuxV, with the key requirement being that Aux asymmetrically c-command the subject at the relevant stage of the derivation. (Recall the case of AuxVV in German in (68).) In a Hindi-type language, the

⁸⁸Notable, too, are Gibson (2013) and Gibson and Marten (2019) on Rangi future tense VAuxO, as well as Hadermann (1996).

⁸⁹Bošković (2016, note 10) envisions the possibility that the limitation in question will turn out just to be a strong tendency.

⁹⁰Silent articles go back at least as far as Chomsky (1965, 107). For detailed discussion, see Longobardi (1994). The existence of silent articles in some languages only in some cases (cf. also Kayne (2008c), Cheng and Sybesma (2012), Jiang (2020)) increases the plausibility of the claim that all languages have articles (vs. Bošković (2009)). For recent discussion, see Crisma and Longobardi (in press) and references cited there; also Köylü (2021).

subject will raise (probably via smuggling) to a position above Aux too soon in the derivation for *have* to be allowed, with this early raising ultimately reflected in Hindi lacking any possibility of AuxXV order, all this from an antisymmetric perspective. (From a symmetric perspective on syntax, no correlation between raising and word order of the sort discovered by Mahajan would be expected.)

Although various details left implicit in the preceding paragraph will need to be filled in, Mahajan's correlation between word order and presence vs. absence of *have* can with considerable confidence be taken to show that what might have been thought to be a purely lexical fact (whether a language has a verb *have* or not) is really in good part a fact about syntax.⁹¹

26. Adjacency effects.

There are VO languages such as English in which V and O cannot be separated by an adverb even when the O is definite (setting aside heaviness effects):

(118) You've read that paper carefully, I see.

(119) *You've read carefully that paper, I see.

What seems to be unattested, however, is an OV language (of the sort that has all the pieces of the VP ending up preceding V) that would systematically forbid its adverbs from intervening between a definite O and a following V. In a symmetric syntactic universe, this asymmetry concerning adverbs with respect to VO vs. OV would be unexpected.

When O is indefinite/bare, the facts are not exactly the same, i.e. there are sometimes adjacency effects, perhaps due to the availability in such cases of incorporation, as Kornfilt (2003; 2018) discusses for Turkish. In fact, adjacency effects of a certain sort are found in English compounds:

(120) Mary plays bridge (well).

(121) Mary is a bridge-(*well-)player.

for reasons that will need to be spelled out, along with an account of the perhaps not very closely related:

(122) *Mary is a well-bridge-player.

We can note in passing that English particles act unlike adverbs in both (119) and (121):

(123) You should read out the figures.

(124) Mary is an avid music downloader.

⁹¹Cf. Harves and Kayne's (2012) argument that the presence in a given language of a transitive verb corresponding to English *need* depends on it having *have* (for later discussion, see Kayne (2016, note 30)).

To be noted is that both Mahajan's idea on *have* and ours on *need* depend in part on how *have* itself is analyzed; the proposal in Kayne (1993) (cf. Freeze (1992)), which used 'late insertion', needs tweaking to be made compatible with Collins and Kayne's (2023) rejection of 'late insertion'. Relevant is the possible decomposition of *have* as *ha+ve*, with *-ve* possibly a P, and *ha-* the same as the non-negative portion of *ain't*. This would maintain *have* as systematically non-transitive, differing then from Myler (2016).

In taking (124) to contrast minimally with (121), I am taking *download* not to be a simple lexical item,⁹² but to display the particle *down* followed by the verb *load* (that together may not even form a constituent), much as in the first of the following two examples, which seem to be interpretively equivalent:

(125) They downplayed the importance of it.

(126) They played down the importance of it.

27. Germanic particles.

Particles (Prt) themselves show a notable asymmetry within Germanic. If one takes the set of Germanic VO languages (English + Scandinavian + associated dialects), one finds both VPrtO and VOPrt, as in English:

(127) Pick up that book!

(128) Pick that book up!

depending on various factors, including the choice of language (Swedish has only VPrtO, Danish only VOPrt).⁹³ Whereas in the Germanic OV languages, the relative order of O and Prt appears to be fixed (even when Prt is 'separable'); one finds (abstracting away from V-2) only OPrtV, never, it seems, *PrtOV.

A specific account of this antisymmetry-friendly fact will require a general understanding of Germanic particles (and comparable particles in other language families) that would go beyond the scope of this essay. Let me just note here the importance of taking into account not only how particles interact with single objects, but also how they interact with double objects, as in the notable discussion in an earlier framework by Emonds (1972), who notes (p. 556) that the most natural position for the particle is as in:

(129) They sent John up a couple of sandwiches.

In addition to *up*, Emonds gives examples parallel to (129) with the particles *out*, *back*, *off*, and *down*. He also gives one with *together* as marginally acceptable:

(130) ?The teacher put the children together the trucks.

This example with *together* is for me close to fully unacceptable, in a way reminiscent in my English of the fact that *away* doesn't work with two objects:

(131) They gave away all their money to their children.

(132) *They gave their children away all their money.

perhaps because *away* is bimorphemic with both subparts (*a+way*) pronounced.⁹⁴

Notable, with regard to (129), is the fact that Norwegian, according to Tarald Taraldsen (p.c.), doesn't allow it (despite otherwise looking a lot like English with respect to particles), for reasons that need to be determined.

⁹²Cf. the separability in Persian of light verb and preceding noun noted by Mohammad and Karimi (1993, 196); and similarly for Korean by Chae (2023, 846); cf. also Cheng (2023).

⁹³For further details, cf. Collins and Thráinsson (1996, sect. 6) and Lundquist (2014).

⁹⁴Cf. Kayne (2016, sect. 3); perhaps *together* can marginally be taken to be *to+gether*.

Of note is the contrast between:

i) ?The Red Cross has been donated millions this year.

on which, see O'Grady (1980, 401), and:

ii) *?The Red Cross has been given away millions this year.

28. Negative phrases.

There is another asymmetry within Germanic, insofar as English appears to be the odd man out when it comes to the positioning of negative phrases such as *no novels*, in particular in the way it differs from Scandinavian languages, which have the property, known within generative syntax going back to Christensen's (1986) and Christensen and Taraldsen's (1989) discussion of Norwegian, that can be seen in the following Norwegian pair:

(133) Jon leser ingen romaner. ('J reads no novels')

(134) *Jon har lest ingen romaner. ('J has read no novels')

As Christensen shows, these examples reflect a general fact about Norwegian negative phrases like *ingen romaner*, which are possible as postverbal objects only if the associated lexical verb has raised in the manner of V-2.

Thinking of Haegeman's (1995, chap. 4) and Haegeman and Zanuttini's (1996, sect. 6.2.4) discussion of West Flemish, a Germanic OV language/dialect, and in particular of their use of a Neg-Criterion, one aspect of which is stated by them as follows:

(135) Each Negative phrase must be in a Spec-head relation with a Neg X°
the unacceptability of (134) can be said to follow from (135), if a phrase in post-participial position cannot be in Spec,NegP. The acceptability of (133) will, from this perspective, reflect the fact that V-2 has placed the finite matrix verb higher than Spec,NegP.

We must now ask why (134) in Norwegian contrasts sharply in acceptability with its English counterpart, as in:

(136) John has read no novels.

Kayne (1998b) has English allowing movement of negative phrases like *no novels* past a participle to Spec,NegP. That movement was then followed by the raising of VP (or some larger constituent) past NegP:⁹⁵

(137) John reads no novels → (neg phrase preposing)

John no novels_i reads t_i → (VP-preposing)

John [reads t_i]_j no novels_i t_j

This VP-movement past NegP in English, in addition to recalling (67), (93), (94) and (116)/(117), looks to be significantly similar to the movement of the complement of Neg⁰ to Spec,NegP proposed by Nkemnji (1995) for the Grassfields language Nweh; it also brings to mind the Dutch dialect with a final negative morpheme mentioned in Haegeman and Zanuttini (1996,162). (That Scandinavian languages cannot mimic the derivation of (137) might have to do with their all being V-2 languages.)

29. More on negative phrases.

⁹⁵In section 3 of that paper I proposed a similar analysis for English *only*, with a key property being that in a sentence like:

i) John criticized only Bill.

only and *Bill* do not form a constituent. If correct, that may weaken the argument given in Carstens and Zeller (2020), as well (as pointed out to me by Gary Thoms (p.c.)) as the one given in Neeleman (2017) against Cinque (2005).

In Kayne (1998b), I proposed that English in some cases allows (overt) long-distance movement of a negative phrase, such as in the following, under the wide scope interpretation of *not a single*:

(138) In all these months, she has suggested that they read not a single linguistics book.

A quick sketch of a derivation would be:

(139) ...suggested that they read not a single linguistics book → (neg phrase preposing)

...not a single linguistics book_i suggested that they read t_i → (VP-preposing)

...[suggested that they read t_i]_j not a single linguistics book t_j

This kind of derivation has the perhaps unexpected consequence that in (138) the embedded subject pronoun *they* ends up not c-commanding the embedded object *not a single linguistics book*. That the unexpected can sometimes be right can be seen as follows. If we take a counterpart of (138) with an 'ordinary' object, we see a Condition C effect in, for example:

(140) Recently, she requested that he revise one of John's articles.

In (140), *he* cannot at all readily have *John* as antecedent. Yet the facts are different in:

(141) In all these months, she has requested that he revise not a single one of John's articles.

where, to my ear, *he* can more readily have *John* as antecedent.

The skeptical reader may have noted that the embedded object in (141) is somewhat 'heavier' than the embedded object in (140). And in fact heaviness is relevant, in addition to negation. Put another way, heaviness by itself can affect Condition C-type judgments:

(142) Recently, she requested that he revise the very one of John's articles that I had been telling you about just last week.

In (142), taking *John* as antecedent of *he* is surely more possible than it is in (140).

(There is no real surprise here, insofar as Heavy-NP-Shift sentences, given the discussion starting at (85), have a derivation rather similar to (137)/(139).)

To close this section on negative phrases, we can note that a 'simple' negative phrase-containing sentence like:

(143) This year, he revised not a single one of John's articles.

maintains a strong Condition C-type violation; *he* in (143) cannot take *John* as antecedent. The reason is that movement of *not a single one of John's articles* to Spec,NegP does not cross the subject position that contains *he*.

Section 5. Sentence-final-ness.

30. The phrasal character of sentence-final particles.

The contrast between (140)/(143) and (141)/(142) concerning Condition C effects is another illustration of a point discussed earlier at (20)/(21), namely, that although hierarchical structure determines linear order unambiguously (given antisymmetry), linear order does not unambiguously determine hierarchical structure. As indicated in the VP-movement based derivation given in (139), we have a derivation-final VO order in (141) (between *revise* and *not a single...*) that looks just like the VO order seen in

(140) (between *revise* and *one of...*); yet these two instances of VO do not correspond to the same hierarchical structure, as the Condition C effect difference confirms.

The VP-movement (or movement of a phrase containing VP) that plays a key role in (139) is similar in certain respects to the movement operations that I would like to briefly discuss in this section. As an initial example, we can think of Pan's (2022) analysis of (different sorts of) sentence-final particles in Chinese as involving leftward movement (to a high position in the left periphery) of the complement of what ends up merely looking like a sentence-final particle.

Pan (2022) takes the movements in question to be of the complement-to-specifier type, and notes (p.151) the tension with Abels's (2003) on antilocality (relevant to which is the proposal in (50) above that complements can never be moved). For present purposes, what is important is simply the existence of such large-phrase movements; whether the landing site is or is not higher than Pan (2022) has it is a separate (if important) question.

There is also the question whether Pan (2022) is correct to take these sentential particles to be simple heads. What I have in mind here is the proposal I made in Kayne (2016) concerning English sentences like:

(144) Where do they live, again?

Although one might be tempted to take *again* here to be a sentence-final particle, I argued in that paper that it might be better to take *again* to be the only pronounced piece of a 'sentence-final sentence', as in:⁹⁶

(145) where do they live, TELL ME again

with the other pieces, TELL and ME, being silent, and with the pre-movement structure looking more like:

(146) TELL ME again where do they live

The preposing of *where do they live*, though compatible with taking *again* to be a simple head, is equally compatible with the analysis sketched in (145)/(146).

Arguably similar to (144) in one key respect are English sentences with initial *only*, of the sort seen in:

(147) Only, I don't want to.

with an interpretation close to that of:

(148) The (only) thing is, I don't want to.

and with an analysis that plausibly involves silent THING (and perhaps silent THE and IS):

(149) THE only THING IS, I don't want to

In (my) English, this *only* does not give rise to preposing (for reasons that remain to be discovered):

(150) *I don't want to, only.

But Pan (2022, 122-3) has an example with a sentence-final particle *eryi* that he paraphrases as 'it is just the case', suggesting the possibility that this *eryi* is accompanied by silent CASE (or perhaps THING), parallel to (149). This partial

⁹⁶This might lead to an account of the fact discussed by Biberauer (2017) and Zeijlstra (2023), namely that particles that are inflected do not produce FOFC violations, if 'clause-final particles' that do produce such violations are in fact all phrasal, and if the phrases in question are incompatible with 'outside' inflection.

alternative to Pan's proposal would not affect his proposed large-phrasal movement; it would, however, essentially eliminate the Spec of *eryi* as the landing site for that movement, in a way that would then be compatible with Abels's (2003) antilocality (and with (50)).

Let me note in passing that a conceivable modification of (146) might be:

(151) CAN YOU TELL ME again where do they live
whose silent YOU, ME and TELL would, even more strongly than (146) itself, recall and support Ross's (1970) performative analysis,⁹⁷ with a further similarity being that this TELL requires being in a root sentence, as seen in:

(152) *I wonder where they live, again.
which contrasts with the possible:

(153) I wonder if you can tell me again where they live.

In further passing, we can note that the proposal in Miyagawa (2022, 18) (and in the work of others that he cites) that uses a SpeakerPhrase and an AddresseePhrase instead of Ross's DECLARE has the property that it inserts *speak* and *address* into the Phrase labels, without there being any clear theory of how complex such a label can be. Ross's (1970) proposal, on the other hand, ties in naturally, not only with (151), but also with other cases of silent predicates, thinking in part of Riemsdijk (2002) on silent GO and in part of silent CAUSE, as proposed by many authors over the years, with a particularly striking example being:⁹⁸

(154) The flowers are on the kitchen on purpose.

31. Partial Wh-Movement and WHAT.

Related to (144) in some respects is:

(155) Where do they live, do you think?
which seems to me to be quite close to what Dayal (2000, 171) calls sequential questions, e.g.:

(156) What do you think? Where do they live?
even though (155) feels like a single sentence and (156) like a pair of linked ones. The partial similarity between (155) and (156) suggests an analysis of (155) in which the embedded sentence *where do they live* is preposed from a position following *think* (much as in (144) and as in Pan (2022)):⁹⁹

(157) [where do they live]_i do you think t_i
and in which, in crucial addition, there is a silent counterpart of the *what* seen in (156):

(158) [where do they live]_i WHAT do you think t_i

The parallellism between (155) and (144) is emphasized by the fact that, just as (144) cannot be embedded, as seen in (152), so, too, is it impossible to embed (155):

(159) *I'm not sure where they live, (do) you think.

The silence of WHAT of (158) is essentially the same (abstracting away from the expletive vs. non-expletive question) as that of the null expletive proposed for Malay

⁹⁷Cf. the possible advantage mentioned in Chomsky (2004, note 17), citing Jon Nissenbaum.

⁹⁸Cf. Kayne (2023), which also proposes a silent causer. On CAUSE, cf. also section 7 above.

⁹⁹Cf. Kayne (1998b, note 107).

Partial Movement in Cole and Hermon (2000, 127) and by Sabel (2000, 441) for Kikuyu. This WHAT in addition recalls Chomsky (1977) on English comparatives, as in:

(160) Mary isn't taller than WHAT she was five years ago.
as well as Svenonius and Kennedy (2006, 144) on (northern) dialectal Norwegian having a null operator in questions such as:

(161) Er du gammel? ('are you old')
with no overt wh-phrase, yet with the interpretation of *How old are you?*, and similarly for Lohndal (2010) on Norwegian exclamatives containing no overt wh-word or other degree word, such as:

(162) For en tulling han er! ('for a fool he is')
which Lohndal argues to contain an unpronounced/silent wh-phrase (WHAT), as in:

(163) HVA for en tulling han er!

Also in part related to (144), I think, is McCloskey's (2000, note 13) West Ulster English example:

(164) He's wile rich but. (= 'but he's very rich')
which lends itself to an analysis in which *he's wile rich* moves to the left past *but*. The same may well be true of standard English:

(165) He's very rich though.
with *he's very rich* moving leftward past *though*.¹⁰⁰

In addition, there are sentences like the following:

(166) A lot of people liked the movie *ø ish*.
These (not part of my own English) are analyzed in depth by Duncan (2016), who argues "that *ish* Merges above the VP/PP it modifies, which then moves past *ish* to derive the surface structure".¹⁰¹

Section 6. Coordination.

32. Some basics.

An antisymmetry-friendly window on the architecture of the language faculty is also provided by a certain type of coordination, as Zwart (2009) argues.¹⁰² According to Zwart, if one looks cross-linguistically at NP/DP-coordination counterparts of English *and*, and if one limits oneself to coordinations in which this counterpart of *and* appears only once, one finds that *and* and its counterparts invariably occur between the two conjuncts:

- (167) a. NP and NP
 b. *and NP NP
 c. *NP NP and

¹⁰⁰Which movement is blocked by *even* and by *al-*:

- i) *He's very rich even though.
ii) *He's very rich although.

¹⁰¹Cf. Mensching's (2015) analysis of certain Sardinian interrogatives, for which he argues "that DPs are extracted to a functional projection beneath FocP before the remnant TP is moved to [spec, FocP]".

¹⁰²Cf. the discussion starting with example (3) above.

Zwart draws the reasonable conclusion that this limitation to one possible order must be reflecting absence of movement. In antisymmetric terms, this seems to be telling us that *and* is a head,¹⁰³ that the two conjuncts are specifier and complement of *and*, and that the order is as it is in (167) because S-H-C order is the only order made available by the language faculty.

Concerning (167b), Zwart is in agreement with Stassen (2000, 15). However, Stassen (2000, 14) notes that (167c) does not hold in Latin, Pitjantjatjara and Tubu.¹⁰⁴ At the same time, Stassen makes the plausible proposal that these apparent counterexamples to (167c) can be thought of as involving deletion of a second ‘and’, i.e. they would in fact be of the ‘NP *and* NP *and*’ type, and hence not true counterexamples to (167c).

A second possible way in which such apparent counterexamples to (167c) might be dissolved would rest on the claim that in (at least some of) the relevant languages, the apparently coordinating element in question does not actually correspond to *and*.

This has some plausibility for Latin *-que*, to judge by Gildersleeve and Lodge’s (1895, 300) statement that Latin *et* “is simply *and*, the most common and general particle of connection” and that “*-que* (enclitic) unites things that belong closely to one another”, which suggests to me that *-que* might be closer to English *as* (cf. *just as*, *as well as*) than it is to English *and*; if so, Latin *-que* would not be relevant to (167c).

The head status of *and* is also suggested by the apparent fact that the ‘*and* NP *and* NP’ type of coordination that is possible in French:

(168) *et Marie et Jean* (‘and M and J’)

is only found in languages that would otherwise informally be thought of as ‘head-initial’; and by the related apparent fact that the ‘NP *and* NP *and*’ type (pointed out for Japanese by Kuno (1973, chap. 8)) is only to be found in ‘head-final’ languages.¹⁰⁵

Indirectly related to this asymmetry is Johannessen’s (1998, 270) comparable discovery that “out of 12 OV languages, 11 have the deviant UC [Unbalanced Coordination] conjunct in the first position; out of 14 VO languages, all have the deviant UC conjunct in the second position”. That coordination is not symmetric is also supported in a number of other ways by Johannessen’s (1998) general discussion of Unbalanced Coordination.¹⁰⁶

¹⁰³Note in particular Oda’s (2017, sect. 5) taking Japanese &P to be head-initial.

The exact implications of Kayne (2016) for the general conclusion about *and* and its counterparts need to be determined, relevant to which is the fact that French allows, in addition to:

i) *ou Marie ou Jean* (‘or M or J’)

the use of the more complex *ou bien*:

ii) *ou bien Marie ou bien Jean* (‘or well M or well J’)

¹⁰⁴Cf. He et al.’s (2017, 5) Pɣnru example in which the nominal coordinator *rəu* follows the last conjunct.

¹⁰⁵Worth noting here is Kroch (2001, 706) on “The lack of consistency in directionality in most languages...”; also Kayne (1994, xv, 143) on Japanese (and Korean) being inconsistent in that sense, based in part on a proposal by Brody (1990, 116) concerning Hungarian *is* (‘also’); cf. also Cinque (2017; 2023).

¹⁰⁶Cf. the fact that Papago/O’odham, to judge by Zepeda (1983, 26), has the property that second sentence of a conjunction of sentences must be non-finite/lack an auxiliary.

33. AND

On a somewhat different tack, let us note Chomsky's (2020, 49) discussion of:

(169) I met someone young, happy, eager to go to college, tired of wasting his time,

...

which he characterizes as having "an unbounded number of possible adjective phrases and no structure among them". To my ear, however, (169), with its '...', is not a sentence of English, as opposed to a closely related example given in Chomsky et al. (2019) containing *etc.*:

(170) John is tall, happy, hungry, bored with TV, etc.

which in turn recalls:

(171) John is tall, happy, hungry, bored with TV, and so on.

(172) John is tall, happy, hungry, bored with TV, and so forth.

It seems plausible to take (170) to contain *and*, just as do (171) and (172), with the twist that in (170) *and* is realized either as silent AND or as the *et* subpart of *etc.*

If so, then (170) in fact reduces to the general question of coordination with *and*. (There is in addition the question of how to characterize the antecedent of *so* (cf. *in that way*¹⁰⁷) in (171) and (172), together with the question whether or not (170) contains a silent counterpart of *so*.) We can now conclude, given all of the previous section, that (170) does have structure and that that structure is not at all symmetrical.

34. Both and either.

The asymmetrical character of coordination shows through in still another way if we take into account:¹⁰⁸

(173) I invited both Mary and John.

together with the impossibility of:

(174) *I invited and Mary both John.

From the perspective of antisymmetry, *both* can introduce only the higher conjunct (perhaps because *both* needs to count as antecedent of *and*). If coordination were thoroughly symmetric, this property of *both* would be more difficult to understand.

We can note in passing that (174) has, against the background of:

(175) I'll invite either Mary or John.

a counterpart with disjunction:

(176) *I'll invite or Mary either John.

so that disjunction is almost certainly no more symmetrical than coordination, with (176) perhaps again reflecting a fact about antecedence, in this case that *either* needs to antecede *or*.

In a partially similar vein, we have:

(177) You do it, or else I'll have to.

(178) *Else you do it, or I'll have to.

with *else* arguably needing to take the first conjunct as antecedent in (177); whereas *else* in (178) cannot, given an antisymmetric approach to disjunction, take the second conjunct, which it asymmetrically c-commands, as antecedent.

¹⁰⁷Recalling Chomsky (2020, 50) on *in that order* (and *respectively*), with much yet to be understood, as he emphasizes.

¹⁰⁸For relevant discussion, cf. Progovac (1999b).

It may very well be the case that this fact about *e/se* is closely related to:

(179) You do it, otherwise I'll have to.

(180) *Otherwise you do it, I'll have to.

with the *other* subpart of *otherwise* matching *e/se*,¹⁰⁹ and the *-wise* subpart matching the *way of in that way*, as mentioned in the brief discussion of *so* in (171) and (172).

Returning to coordination proper, we have:

(181) Mary left early and John did, too.

(182) *Mary left early, too and John did.

As with *else* and *otherwise*, *too* needs an antecedent, which it can find in the first (asymmetrically c-commanding) conjunct in (181), but not, given an antisymmetric analysis of coordination,¹¹⁰ in the second (asymmetrically c-commanded) conjunct in (182).

Arguably akin to (181)/(182) is the following pair:

(183) ?Mary left early and, in addition to that, John did.

(184) *Mary left early in addition to that, and John did.

with *that* in (183) taking the first conjunct as antecedent, and with *that* in (184) unable to take the second conjunct as antecedent for by now familiar reasons of c-command.

Its similarity to (183) suggests that (181) might be analyzed as:

(185) M left early and J did IN ADDITION to THAT

with *too* now seen to be the same element as *to* despite the orthographic difference, in a way that would give support to the antihomophony principle/heuristic of Johns (1992, 84), Embick (2003, 156) and Kayne (2019a, 137).

35. Idioms and the asymmetry of coordination.

Idioms of a certain sort provide an additional window on the asymmetry of coordination. Consider:

(186) This wine is nice and cool.

(187) This wine is cool and nice.

Of these, (186) lends itself very naturally to an interpretation close to that of:

(188) This wine is nicely cool.

whereas (187) does not. Similarly for:

(189) Make it good and hot.

(190) Make it hot and good.

where (189) has a natural interpretation close to that of:¹¹¹

(191) Make it really hot.

while (190) does not.

The generalization would seem to be that an adjective in the first conjunct (but not one in the second conjunct) can in some cases idiomatically take on an adverb-like interpretation.¹¹² If the first conjunct asymmetrically c-commands the second, these

¹⁰⁹On *other* and *else*, Kayne (2022b; 2022c) and references cited there.

¹¹⁰Which may, along with an antisymmetric analysis of disjunction, be relevant to Schwarz (2013).

¹¹¹Cf. French *bien chaud* ('well hot').

¹¹²How close these are to Wiklund's (2007) 'pseudocoordinations' remains to be determined.

facts can be related to the fact that adverbs typically asymmetrically c-command what they modify, as in Cinque (1999). If the two conjuncts c-commanded each other, we would have difficulty understanding the asymmetries in (186)-(190).

Section 7. Pronouns and their antecedents.

36. Pronouns and their non-c-commanding antecedents.

English readily allows both of the following:

(192) The fact that John is here today means that he's well again.

(193) The fact that he's here today means that John is well again.

Both (192) and (193) have the property that in them neither *John* nor *he* c-commands the other, with English thereby giving the impression that in such non-c-command configurations anything goes. This impression may have fed into Lasnik's (1976) claim that pronouns could in general freely take antecedents subject only to (what later came to be called) conditions B and C of the binding theory.¹¹³

It appears, though, that English is not fully representative, insofar as various languages allow sentences like (193) less readily than English, or not at all. For example, Michel DeGraff (p.c.) has told me that in Haitian Creole "backward pronominalization"¹¹⁴ of the sort seen in (193) is systematically impossible. In addition, Huang (1998, sect. 5.5.2) indicates that Chinese has much less backward pronominalization than English. And Craig (1977, 150) in her grammar of Jacaltec says that Jacaltec has no backward pronominalization at all. Strikingly (since Danish is a Germanic language like English), Allan et al.'s (1995, 473) grammar of Danish says that Danish has either none or at least much less backward pronominalization than English (cf. Thráinsson et al. (2004, 331) on Faroese, another Germanic language).¹¹⁵ Finally, Jayaseelan (1991, 76) says about Malayalam that some speakers of Malayalam allow no backward pronominalization at all.

In contrast to these sometimes severe limitations on backward pronominalization found in some languages, I don't know of any language that in a systematic way completely or even partially prohibits forward pronominalization of the sort seen in (192). There thus seems to be a precedence-based asymmetry concerning antecedent-pronoun relations in contexts of non-c-command, of a sort that would be unexpected in a symmetric syntactic universe.

A proposal to account for this asymmetry was made in Kayne (2002). Oversimplifying a bit, the proposal was in essence that the language faculty requires that the antecedent of a pronoun must precede that pronoun at some point in the derivation.¹¹⁶ Thus in (193), there must be a stage in the derivation at which *John* precedes *he*, and there must apply in the derivation of (193) a movement operation (or more than one) that accounts for *he* subsequently coming to precede *John*.

¹¹³On the idea that neither Condition B nor Condition C is primitive, see Kayne (2002).

¹¹⁴This term goes back to the 1960s; cf. Langacker (1969).

¹¹⁵Cf. also Antonyuk and Bailyn (2008) on Russian.

¹¹⁶Cf. Bruening (2014). Kayne (2002, section 7) explicitly took discourse to be a subtype of coordination (cf. Hoekstra (1999)), in order to integrate cases like:

i) John is famous. He's smart, too.

The expectation then is that languages with no backward pronominalization will not have access to that type of derivation. Languages with some backward pronominalization (but less than English) will have only limited access to the relevant type of derivation.¹¹⁷

37. Precedence must be visible to semantic interpretation.

Since the antecedent-pronoun relation is of importance to interpretation, the fact that it is cross-linguistically sensitive to precedence, as just discussed, means that, at the very least in this case, precedence must be visible to interpretation, and therefore must be part of core syntax. (That precedence is part of core syntax will come up again later.)

38. Weak crossover.

That the antecedent of a pronoun must precede that pronoun at some point in the derivation is a necessary, but not sufficient condition for well-formedness. A well-known interpretively sensitive example is that of weak crossover,¹¹⁸ as in the following, which contrasts with (193):

(194) The fact that he's here today means that every little boy is well again. Although acceptable with *he* taking as antecedent a phrase contained in a preceding sentence or discourse, (194) is hardly acceptable with *he* taking as antecedent *every little boy*. From the present perspective, this fact is arguably related to the fact that the following sentence also strongly resists taking *every little boy* as the antecedent of *he*:

(195) The fact that every little boy is here today means that he's well again.

Part VI. Linear order and externalization

39. Linear order is part of core syntax.

The conclusion reached in section 37, to the effect that at least sometimes precedence must be visible to interpretation, is at odds with the suggestion that Chomsky (2019; 2020) has made in recent work, when he says that linear order is not part of core syntax and that linear order comes into play only as the result of what he calls externalization. Let us now ask to what extent antisymmetry might or might not be compatible with Chomsky's view of externalization.

Antisymmetry claims that linear order is dependent on structure; in particular for every projection, the mapping to linear order must invariably yield Spec-Head-Complement order, whatever the type of projection. There is no optionality of the head-parameter sort. Antisymmetry as in Kayne (1994) also required that there be only one Spec per projection, in a way that has fed into cartography work, as exemplified by Rizzi (1997) and Cinque (1999). My impression is that from Chomsky's externalization perspective one could, if one agrees that S-H-C is correct, in fact incorporate antisymmetry into the mapping from core syntax to PF.

Why, though, does antisymmetry hold to begin with? (In 1994 terms, why is the LCA part of UG?). In Kayne (2011), I suggested that (the beginning of) an answer to this

¹¹⁷What kind(s) of parameter(s) underlie these differences remains to be determined.

¹¹⁸Cf. Postal (1971) and much subsequent work.

question is available, but that it requires, as in section 37, taking linear order to be part of core syntax. More specifically, the suggestion was to make use of an alternative to standard Merge that was mentioned but not pursued in Chomsky (2008),¹¹⁹ namely that Merge should always be taken to form the ordered pair $\langle X, Y \rangle$, rather than the set $\{X, Y\}$, with that ordering necessarily to be understood as temporal ordering.

When we say, for example, that English prepositions precede their objects, we are saying that prepositions are pronounced prior to the pronunciation of their object. Sentences and discourse unfold in time. (Having said this, I will for the most part keep to the term 'linear order', returning to 'temporal order' and the related question of time later.)

Chomsky (2020) takes the opposite view from that of Kayne (1994; 2011), i.e. he takes the view that linear/temporal order is not at all part of core syntax.¹²⁰ In part he does so on the basis of the point that differences in linear order do not feed differences in semantic interpretation, which depends only on structure (and on lexical items and features).

But the force of this point rests on the assumption, denied by antisymmetry, that linear order can vary independently of structure (as it could in the head parameter tradition). If, on the other hand, linear order is fully determined by hierarchical structure, then there is no reason to expect it to be able to make its own independent contribution to semantic interpretation. If that is so, then linear order can, as far as interpretation is concerned, perfectly well be part of core syntax.

40. More on linear order as part of core syntax.

Another reason why Chomsky has taken linear order not to be part of core syntax has to do with examples that show that internal merge (movement) cannot take the linearly closest auxiliary in English subject-aux inversion.¹²¹ Thus starting from:

(196) Somebody who is in Paris is on the phone.

one cannot conceivably derive:

(197) *Is somebody who in Paris is on the phone?

The question arises, though, as to whether this strong prohibition might (or might not) derive from independent structural factors, e.g. from the general impossibility of extraction from within a subject phrase, or from within a relative clause. Relevant, needless to say, is the fact that extraction from within a subject phrase or from within a relative clause does not always yield a violation as strong as that of (197). For example, to my ear the following instance of extraction from within a subject is less sharply deviant than is (197):

(198) ??He's somebody whose close friends of generally like linguistics a lot.

The difference in degree of acceptability, as compared with (197), is stronger with parasitic gaps:

(199) ?He's somebody whose close friends of generally admire.

¹¹⁹Cf. Chomsky (2020) on Pair-Merge, though his use of it is different from mine, as is Saito and Fukui's (1998), which retains a head parameter orientation.

¹²⁰Cf. Chomsky (1995, 340).

¹²¹Indirectly relevant here is the question whether Internal Merge is triggered; for Chomsky (2019, 268) it is not.

One can, on the other hand, reach a violation as sharp as that in (197) if, instead of extracting an argument from within a subject, as in (198) and (199), one tries to extract a non-argument such as an adverb. Thus, starting from:

(200) Somebody who was speaking loudly left very suddenly.

it is sharply impossible to derive:

(201) *How loudly did somebody who was speaking leave very suddenly?
and similarly even for parasitic gaps (where the notion of 'linearly closest' is not obviously relevant). For example, starting from:

(202) Somebody who was behaving badly was near somebody else who was behaving badly.

one cannot reach:¹²²

(203) *How badly was somebody who was behaving near somebody else who was behaving?

Similarly, a parasitic gap counterpart of (197) remains very strongly deviant:¹²³

(204) *Is somebody who in Paris on the phone?

The same set of points can be made with regard to extraction from within a relative clause. As Chung and McCloskey (1983, 708) noted, English does allow such extraction in some cases, e.g.:¹²⁴

(205) That's one trick that I've known a lot of people who've been taken in by. However, this example, and all the others they cite (as being "either well formed or much more acceptable than one would expect"), are instances of the extraction of an argument.

Examples similar to any of theirs, but involving the extraction of an adverb, are invariably sharply deviant, as far as I can see. For example, alongside:

(206) We've known a lot of people who word their messages carefully.
there is no:

(207) *How carefully have you known a lot of people who word their messages.

The strong deviance of (201), (203), (204) and (207) is independent of any notion of linear closeness. Consequently, the strong deviance of (197), if assimilable to that of (201), (203), (204) and (207) in terms of structural constraints on the extraction of non-arguments, can be understood in a way that is fully compatible with linear order being part of core syntax.

41. Counting is not part of syntax.

Chomsky (2020) also mentions an aspect of work by Moro et al. (2003) showing that Broca's area activation does not take place when subjects are presented with an 'unreal' language in which, for example, negation would systematically be the third word in a sentence. This is a telling point, but I don't think it bears directly on the question

¹²²The same holds of Chomsky's (2020; 2022, 352) example:

i) Carefully the guy who fixed the car packed his tools.

The adverb *carefully* cannot originate within/be extracted from the relative clause embedded within the subject, whether there is in addition a parasitic gap or not.

¹²³Cf. the fact, discussed by Källgren and Prince (1989, 55), that finite V-movement out of a relative clause island in Yiddish is impossible, even though it leaves a copy.

¹²⁴Cf. Taraldsen (1981) on Norwegian, as well as Bolinger (1972).

whether linear order is part of core syntax. Rather, what it shows, I think, is that the language faculty doesn't count numerically. Clearly the language faculty doesn't count words in linear order.

But it also doesn't numerically count structural notions such as depth of embedding. As far as I know, no syntactic operation takes, or could possibly take, as its goal a phrase that would be exactly three nodes down from the (node immediately dominating the) probe. Nor could any syntactic operation search for a phrase that is the third closest, structurally speaking.

That numerical counting is not countenanced by the language faculty is itself something that needs to be understood (in particular against the background of Chomsky's (2019, (text to) note 27) suggestion that arithmetic is an offshoot of the language faculty), but its unavailability does not imply that linear order is not part of core syntax.

42. Core syntax and semantic interpretation.

Let us now return to the question of semantic interpretation. I have taken the position that linear order doesn't need to directly feed into semantic interpretation, since linear order is antisymmetrically determined by hierarchical structure, which does feed into it. Yet linear order is part of core syntax. Let me take all this to be part of a broader question, namely whether or not semantic interpretation must take into account all aspects of core syntax.¹²⁵

This broader question ties into one raised by Chomsky et al. (2019), who state that "Whether...semantically vacuous scrambling, extraposition, clitic movement etc., ...reflect narrow-syntactic computations or are part of the mapping to PHON...is an open question".¹²⁶

If (53), repeated here:

(208) Arguments invariably raise at least once.

is correct, then even arguments that appear in their canonical/neutral position in the language in question (including in VO languages) will have been moved there in what appears to be a semantically vacuous way (unless the semantics pays important attention to the link between even an argument in canonical position and its trace).

¹²⁵If semantic interpretation ignores some aspects of core syntax, that will have something in common with phonetic interpretation not requiring that all aspects of phonological structure have a phonetic counterpart, e.g. deleted/silent phonological segments, with one example being abbreviated words, as in *abs* (for *abdominal* (*muscle*)s), whose lax vowel (in my English) plausibly reflects the presence of silent DOMINAL at the relevant point in the derivation, contrasting phonetically thereby with *tabs*.

¹²⁶Relevant here is Chomsky's (1995, 151) Full Interpretation principle, whose interaction with idioms calls for clarification. Note also Matushansky's (2006, Appendix) arguments "against analyzing head movement as a phonological phenomenon"; and similarly, Hosono (2019). Also relevant here is Arano's (2022, sect. 4.2) recent argument that scrambling in Japanese must take place in narrow syntax; cf. Saito and Fukui (1998, 445) on heavy-NP-shift (and scrambling). As well as Cinque's (2023, 102) discussion of 'meaningless movements'.

43. VO order in non-finite environments in Estonian and Finnish.

But before I go any further into the question of diagnosing core syntax, let me indulge in a digression concerning (208) and canonical order, via a striking observation due to Vilks (1998), who states that Estonian and Finnish have verb-final order more consistently in non-finite adjuncts than in non-finite complements. Put conversely, those two languages have VO order more consistently in non-finite complements than in non-finite adjuncts. Now Brattico and Huhmarniemi (2006, 10) take the basic word order of Finnish to be SVO.¹²⁷ The question, then, is why the prevalence of this basic/canonical VO order should be at all sensitive, in non-finite contexts, to whether the non-finite embedding is a complement or an adjunct.

If Finnish VO directly reflected external merge positions, contrary to (208), this sensitivity would be very difficult to understand. Whereas if (208) is valid, the likelihood of an account is greater.

That Finnish supports (208) seems clear. As usual, details need to be filled in. What comes to mind is this: VO order in nonfinite clauses in Finnish (and Estonian) is derived via a notable instance of scrambling. More specifically, the O in such cases must be long-distance scrambled out of the nonfinite clause entirely, to a position high up in the matrix. Subsequent to this, remnant movement of the matrix VP (which will include the nonfinite clause) takes the matrix VP to a still higher position.¹²⁸ The reason that VO order in non-finite clauses is less readily available in adjuncts now reduces to the fact that extraction (of O) out of adjuncts is less readily available than is extraction out of complements.

44. Diagnosing core syntax.

Let us now return to the question of types of evidence that bear on being part of core syntax or not. I will touch on cases in which semantically vacuous syntactic operations can be diagnosed as being part of core syntax indirectly, through their interactions with other core syntactic operations. Take, for example, relative clause extraposition, which feels semantically neutral in pairs like:

(209) Somebody who I used to know in high school just walked in.

(210) Somebody just walked in who I used to know in high school.

Yet there is a restriction that comes to light in the following pair:

(211) The only person who I liked in high school just walked in.

(212) *The only person just walked in who I liked in high school.

presumably due to a property of the scope of *only* that must surely be (keyed to) part of core syntax. Put another way, relative clause extraposition is not available in (212) for syntactic/semantic reasons that cannot possibly be part of PF, even if relative clause extraposition itself is in general interpretively neutral.

¹²⁷Cf. Holmberg (2016, 27).

¹²⁸Cf. the derivations given in Kayne (1998b) for the matrix scope negation reading in English sentences such as those given by Klima (1964), e.g.:

(i) John will force his daughter to marry no one.

See also Bayer (1993).

This point concerning relative clause extraposition harks back to Holmberg's (1986, sect. 6.2) argument that Scandinavian Object Shift is syntactic, i.e. part of core syntax, in current terms. This is so in particular, he argues,¹²⁹ because Object Shift feeds topicalization, itself obviously part of core syntax.

In a partially similar way, consider clitic movement of the Romance type, which in simple cases seems to have no semantic effect. Yet we have contrasts in French of the following sort:¹³⁰

(213) *Combien* (**en*) *ont lu ton livre?* ('how-many (of-them) have read your book')

(214) *Combien tu* *(*en*) *a lus?* ('how-many you (of-them) have read' = 'how many of them have you read')

When one fails to pronounce the noun that goes with *combien* ('how many'), the clitic *en* ('of them') is obligatorily present in the object case (214),¹³¹ yet impossible in the subject case (213). The obligatoriness in (214) tracks the obligatoriness of *en* in:

(215) *Tu* *(*en*) *a lu beaucoup.* ('you (of-them) have read many')

suggesting strongly that the movement of clitic *en* in (214) must be taking place prior to wh-movement. If so, then, since wh-movement is part of core syntax, so must be clitic movement (at least in this case), despite its semantic neutrality.

That clitic movement is part of core syntax is also suggested by its interaction with raising to subject position, in cases in French such as:¹³²

(216) *Le premier chapitre semble en être intéressant.* ('the first chapter seems of-it to-be interesting')

The clitic *en* ('of it') originates within the DP containing *le premier chapitre* ('the first chapter'). It clearly must, in (216), move to clitic position within the embedded infinitival phrase prior to the raising to matrix subject position of *le premier chapitre*. Since that raising is part of core syntax, so must the movement of *en* be, despite its (apparent) semantic neutrality.¹³³

Finally, take verb raising of the sort discussed in Pollock (1989), which as Chomsky (2020) emphasizes is also semantically neutral. Yet in Kayne (1991) I argued that Romance infinitival verb raising has an effect on whether or not PRO is admissible as the subject of a Romance infinitival *if*-clause. In some Romance languages it is, and in others it is not, and this correlates with whether or not the Romance language in

¹²⁹Cf. also Holmberg and Platzack (1995, chapter 6) and Holmberg (2005).

¹³⁰Cf. Kayne (1975, sects. 2.19, 4.3), Rizzi (1982, 148ff.) and Pollock (1998).

¹³¹In a resumptive pronoun-like fashion - cf. Kayne (2022a).

¹³²Originally discussed by Ruwet (1972).

¹³³Relevant here is Uriagereka (2000) on the semantics of clitic doubling; also Déprez (1998) and Obenauer (1992) on semantic effects of past participle agreement in French; as well as Ikawa (2022) on how Agree feeds interpretation with Japanese honorifics.

question moves its infinitives past the landing site of its object clitics.¹³⁴ If so, then Romance infinitival verb raising must, since it bears on the admissibility of controlled PRO, be part of core syntax, despite apparently being semantically neutral.

45. Partial linear order.

The position that I took in section 39, to the effect that Merge should always be taken to form the ordered pair $\langle X, Y \rangle$, rather than the set $\{X, Y\}$, introduces linear order into core syntax, but only in a partial way, in the following sense. The formation of the ordered pair $\langle X, Y \rangle$ tells us that X is linearly/temporally ordered before Y, but at the same time it tells us nothing explicitly about subconstituents of X or of Y. In particular, it does not explicitly tell us that subconstituents of X are linearly ordered before subconstituents of Y.

Put another way, the Merge-based building up of syntactic structure via the formation of ordered pairs creates in core syntax a partial, but not a total, linear ordering. In a way reminiscent of Kayne (1994, 5), a total ordering can be reached by adding:¹³⁵

(217) For all $\langle X, Y \rangle$, all the terminals dominated by X precede all the terminals dominated by Y.

Whether or not we would need to take (217) to be specific to the language faculty is not entirely clear. Nor is it clear, if (217) is specific to the language faculty, whether it would better be associated with externalization in Chomsky's sense, or directly with core syntax. In any event, a partial linear order will suffice, if Kayne (2011) is on the right track, to account for the fact that syntax is antisymmetric (in 1994 terms, to account for why the LCA holds).

¹³⁴This correlation fits in with Kato et al.'s (2023, 309) observation that European Portuguese prohibits clitic-infinitive order with its counterpart of 'if'+infinitive:

i) Não sabemos se {*lhe dizer /dizer-lhe} a verdade hoje ou amanhã. ('neg we-know if him to-tell/to-tell him the truth today or tomorrow')
despite allowing clitic-infinitive order elsewhere. Whether the proposal in Kayne (1991) can extend to comparable facts found in EP with certain prepositions remains to be worked out.

¹³⁵Rizzi (2018, 343) notes the existence of (weaker, as compared with hierarchical) linear intervention effects. The weakness of these linear intervention effects may be due to their being dependent on the precedence relations induced by (198).

Adjacency might appear to be relevant to the type of agreement found in (cf. Kayne (1998a, note 9) and den Dikken (2001)):

i) ?The identity of the participants are to remain a secret.
But this seems unlikely, since the following does not feel very different:
ii) ?The identity of the participants from Boston are to remain a secret.
In addition, my English allows:
iii) ?The participants' identity are to remain a secret.
but not:
iv) *Their identity are to remain a secret.

indicating that this type of agreement reflects (unfamiliar, non-standard) English syntax, rather than just 'errors' as in Bock and Miller (1991).

46. Linear order as temporal order, or, The integration of time into core syntax.

Returning to the question of 'linear order'/'temporal order', we now see that to say that (partial) linear order is part of core syntax is to say that (partial) temporal order is part of core syntax. Conversely, to say that linear order is not at all part of core syntax is to say that temporal order is not at all part of core syntax. Which is in turn to say that the language faculty has not integrated into its core any notion of time.

Put another way, the question for future work (not limited to linguistics) is whether or not the human brain has or has not entirely disregarded time at the very core of one of its most important components, as well as to what extent the human brain integrates notions of time in other areas of cognition, broadly interpreted.

The proposal made earlier (with Merge forming ordered pairs) amounts in effect to saying that the language faculty has met the challenge of externalization by partially integrating temporal order into its core.

Appendix. The Relevance of Phonology

1. Bundling

Let me begin with a digression into the notion of bundling, as the term is used when speaking of the bundling of syntactic features into a lexical item. This notion of bundling is partially incompatible with a proposal made in Kayne (2005a, Appendix) and called there a principle of decompositionality, which read as follows:

(218) UG imposes a maximum of one interpretable syntactic feature per lexical item.

If we strengthen (218) by dropping the word 'interpretable', as in:¹³⁶

(219) UG imposes a maximum of one syntactic feature per lexical item.

then we have a decompositionality principle that in effect says that there is no bundling of syntactic features into a single head.¹³⁷ But if that is the case, the question is, why not? That is, what would lead the language faculty to have the property expressed in (219)?

The answer that I would like to propose runs as follows. Taking a cue from key discussions over many years with Chris Collins and thinking of Agbayani and Ochi (2014) and references cited there, bundling must be an instance of Merge.¹³⁸ Now let me add to this the idea from Kayne (2011, sect. 4), to the effect that every instance of Merge must be associated with a precedence relation. If so, then bundling (= merging) two syntactic features together must result in one preceding the other, in a way that wouldn't fit with the usual interpretation of the term lexical item.¹³⁹

2. Merge and phonology.

¹³⁶Thinking in particular of Shlonsky (1989) on the splitting of agreement features into separate heads, extending the line of thought of Pollock (1989).

¹³⁷Cf. Caha (2015); also Kayne (1994) on the absence of multiple specifiers, and Cinque and Rizzi (2010).

¹³⁸Cf. Chomsky (2001, 10), going back in part to Chomsky's (1995, sect. 4.4.4) 'move feature' idea.

¹³⁹Though there is a point in common here with Starke (2009).

The assimilation of bundling to merge can arguably be extended to phonology. Consider the fact that morphemes are not atomic, insofar as they are composed of (a syntactic feature associated with) phonological segments,¹⁴⁰ in turn composed of phonological features. Assume now that composition, in this sense, is also not distinct from merge.

If so, then the language faculty will have phonological features merging to form segments, and segments merging to give the phonological form of a morpheme, which would then merge with that morpheme's syntactic feature.¹⁴¹ That would mean that phonology is syntax-like in a significant way.¹⁴² Furthermore, we might take there to be just one single 'merge engine', such that bottom-to-top bare phrase structure-type derivations (as in Chomsky (1995, 249)), in fact start with phonological features, continue on up through the phonology and only then reach the syntax. (A resolutely bottom-to-top derivation of the sort that starts with the phonology would preclude late insertion.¹⁴³) If the language faculty has a single merge engine in the preceding sense, then that constitutes a solution to the externalization problem discussed, for example, by Chomsky (2009, 386).

In effect, from this perspective, phonology feeds syntax,¹⁴⁴ and should not be factored out of it.

¹⁴⁰Some morphemes may be monosegmental.

Kayne (2016) argues that syntactic features are linearly separated from the associated phonology.

¹⁴¹I set aside the question of semantic features; for relevant formal discussion, see Collins and Stabler (2016).

¹⁴²Cf. in part the Government Phonology subtradition of the 1980s, as exemplified by Kaye, Lowenstamm and Vergnaud (1990); also Pöchtrager (2006; 2024) and references cited there, as well as Dobashi (2003), Nevins (2010), Kayne (2016) and especially den Dikken and van der Hulst (2020)); for an enlightening discussion of the syllable, see Goldsmith (2014).

I leave open the question of possible mismatches between phonological and syntactic constituency, as in Myers (1987), as well as the question of autosegmental phonology.

The kind of mismatch involving relative clauses discussed by Chomsky and Halle (1968, 372) in terms of readjustment rules rests on the assumption that in the syntax the relative 'head' and relative clause can form a derivation-final constituent; for discussion questioning that assumption, suggesting in effect that relative clause 'extraposition' always holds (with relatives of the postnominal sort), see Kayne (1998a), bearing on Taraldsen (1981).

¹⁴³Both in the DM sense, as in Halle and Marantz (1993; 1994), and in the sense of nanosyntax, as in Starke (2009). Also precluded will be late insertion in the manner of generative semantics and related work, as in, for example, Gruber (1965; 1967, 939), McCawley (1968), Geis (1970, 19ff.), Postal (1970) (on which, see Ronat (1972)) or Lakoff (1971), with the alternatives arguably involving silent elements of the sort discussed in Kayne (2005b).

¹⁴⁴Contrary to the Y-model that has syntax feeding phonology, but in a way that converges with Magrassi et al. (2015). (Nor should there be any 'post-syntactic' operations - for relevant discussion, see Collins and Kayne (2023).)

3. Merge and phonological features.

Given that (temporal) precedence is an integral part of phonology, we can conclude, if merge encompasses in a uniform way both phonology and syntax, that the antisymmetry-based claim that precedence is an integral part of (narrow) syntax is reinforced.¹⁴⁵

If phonological features are brought together by merge, and if it holds with complete generality that the output of merge is, as in Kayne (2011, sect. 4.2), associated with precedence/temporal order,¹⁴⁶ then the expectation arises that phonological features within a segment must be temporally ordered (in addition to being hierarchically arranged¹⁴⁷). Such ordering has actually already been suggested for particular cases, by terms like ‘prenasalized stop’¹⁴⁸ or ‘prestopped nasal’¹⁴⁹, with the first indicating that the nasal feature precedes the feature(s) responsible for the stop character of the segment, and the second the reverse.¹⁵⁰

In a similar vein, Elizabeth Zsiga (p.c.) has noted the existence of post-aspirated stops, while pointing out at the same time that precedence seems to be inappropriate for many pairs of features. On the other hand, she has also noted that voicing often starts ‘late’,¹⁵¹ which from the present perspective might indicate that (in such cases)

¹⁴⁵The contrary idea, namely that precedence should be factored out of syntax, goes back to the advent of X-bar theory in Chomsky (1970).

¹⁴⁶Although the elements directly merged with one another have a temporal order imposed on them, in that proposal, no ordering statement is comparably given that involves subparts of either of those elements. In other words, if a complex phrase ‘[_A C D]’ is merged with B, then either AB or BA must hold, but nothing is immediately said about CB or BC; this may bear on questions of intervention effects (on which, see Rizzi (2009)).

¹⁴⁷From the present perspective, we would expect antisymmetry to be respected both intrasegmentally and intersegmentally. The latter case will tie in to questions of syllable structure, for which ‘onset-nucleus-coda’ may match the linear ‘specifier-head-complement’ arrangement of Kayne (1994; 2011); in addition, the antisymmetry-based prohibition against multiple Specs would recall phonologists speaking of complex onsets, rather than of multiple onsets.

¹⁴⁸Cf. Maddieson (1989); also Chomsky and Halle (1968, 316) and references cited there.

¹⁴⁹Cf. Turpin et al. (2014) and Round (2014).

¹⁵⁰Taken together, these two cases would seem to constitute an example of identical features yielding distinct segments by virtue of distinct segment-internal temporal order.

¹⁵¹Cf. Hock (1991, 121) on “aspiration as delayed voicing onset”; also the very notion of an affricate as a single segment consisting of a plosive followed by a fricative. Hayes’s (2009, 79) discussion of a ‘delayed release’ feature (cf. Chomsky and Halle (1968, 318)) may belong here, too (even if that feature were to turn out to be too complex to be a plausible primitive).

the place of articulation feature(s) must precede the voicing feature.¹⁵²

4. More on merge and phonology.

A counterpart for phonology of Chomsky's (1995, 225ff.) notion of numeration would presumably just be those features and segments that lead up to the lexical items chosen. It would seem that the most straightforward interpretation of phonological numeration would not include epenthetic segments. If so, then Chomsky's (1995, 228) inclusiveness condition, extended to phonology, would arguably exclude epenthetic segments completely, i.e. would prohibit recourse to phonological epenthesis. This would have the desirable effect of shrinking the space of possible analyses that the phonological learner has to cope with; for example, the learner of Polish would not need to (or be able to) entertain an epenthesis-based account of Polish jers, and would be therefore be obliged to adopt an analysis involving deletion of the sort discussed (along with epenthesis) by Hayes (2009, chap. 12).¹⁵³

In a different vein, Dominique Sportiche (p.c.) has pointed out that if bottom-to-top derivations start with phonological features, then we might expect the interpretive component to see such features, just as it sees higher level constituents. Although discussion is well beyond the scope of this paper, this might lead to a possible integration into grammatical theory of the tradition of phonetic symbolism.¹⁵⁴

5. Phonology, syntax, and cyclic rule application.

In Chomsky's (1965, 143) words, "Given a generalized Phrase-marker, we construct a transformational derivation by applying the sequence of transformational rules sequentially, "from the bottom up"". This principle of the transformational cycle, while arguably correct,¹⁵⁵ was not otherwise grounded in the architecture of UG. The advent of "bottom-to-top" derivations in Chomsky (1995) grounded this cyclic principle more deeply. Transformational rules, as instances of internal merge, proceed "from the bottom up" because all instances of merge, both internal and external, proceed in that way.

¹⁵²The question whether and if so how multiple place or manner of articulation features can be integrated is left open. How many features there are will depend in part on whether or not Hayes's (2009, 91) zero valued features can be reinterpreted as absent, in relevant cases.

The features in question are phonological; they presumably impose boundary conditions on the phonetics, without themselves determining every last detail of the phonetics.

On the assumption that signed languages are strongly akin to spoken ones, we would expect the present discussion to carry over to the phonology (and syntax) of signed languages, at some suitable level of abstraction.

¹⁵³For an argument against *r*-epenthesis in Japanese, see Pellard (2016). Homorganic glide epenthesis of the sort suggested by Hayes (2009, 172) might be reinterpretable as involving in part feature movement.

¹⁵⁴Cf., for example, Shrum and Lowrey (2007) and the many references cited there.

¹⁵⁵For one extended argument based on French, see Kayne (1975).

The syntactic transformational cycle of Chomsky (1965) had a parallel in the phonological transformational cycle of Chomsky and Halle (1968, 15). But there, too, one could have asked why phonological rules would happen to apply in precisely that cyclic way.¹⁵⁶ A possible answer from the present perspective is the same as the one just given for syntax. This answer requires that phonological rules or operations be interpretable as instances of internal merge (including Agree, as in Nevins (2010, 192)),¹⁵⁷ in which case their bottom-to-top character will fall out of the bottom-to-top character of phonological merge in general, including external merge (both of features and of segments).

The notion of phonological cycle here will need to be fleshed out in terms of phases.¹⁵⁸ If properly done, that might tell us why standard syntactic movement operations are insensitive to phonology, i.e. they appear never to ‘see’ the phonology. (For example, no syntactic fronting operation picks out phrases whose initial phonological segment is a glide.) The reason might be that, once the point of application of these syntactic operations is reached, the phonology will be too deeply buried, phasally speaking. Alternatively (or in addition) there may be a link to Fodor’s (2009, 202) point, if correct, that there is no syntactic fronting of specifically plural noun phrases; that is, it may be that not even all morphosyntactic features are accessible to (A-bar) movement.¹⁵⁹

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¹⁵⁶This paragraph assumes the existence of phonological rules; for recent discussion of the rules vs. constraints question, see Odden (2014).

¹⁵⁷I agree with Nevins that Bromberger and Halle (1989) took the differences between phonology and syntax to be greater than they are. (Whether extrinsic phonological rule ordering is needed is an orthogonal question.) Internal merge with movement (i.e. not limited to Agree) might be appropriate for metathesis, for infixation and/or for non-concatenative morphology. Relevant in a general way is Chomsky’s (1995, sect. 4.4.4) ‘move feature’ idea. Questions of locality arise - cf. Nevins (2010).

¹⁵⁸On phases, see Chomsky (2001; 2008).

¹⁵⁹Fodor also points out that number sensitivity is, on the contrary, found in:

- i) How tall a man is he?
- ii) *How tall men are they?

Possibly, (ii) is excluded parallel to:

- iii) *How tall one man is he?

with the silent plural determiner in (ii) behaving as *one* in (iii), and with *a* in (i) being less complex than either of these; for relevant discussion, see Kayne (2017b).

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